

DA40 NG



Handout

DA40 NG Systems

Version 5.2

INTENTIONALLY BLANK

Diamond DA40 NG



1

Dimensions



DA40 NG

2

Dimensions



DA40 NG

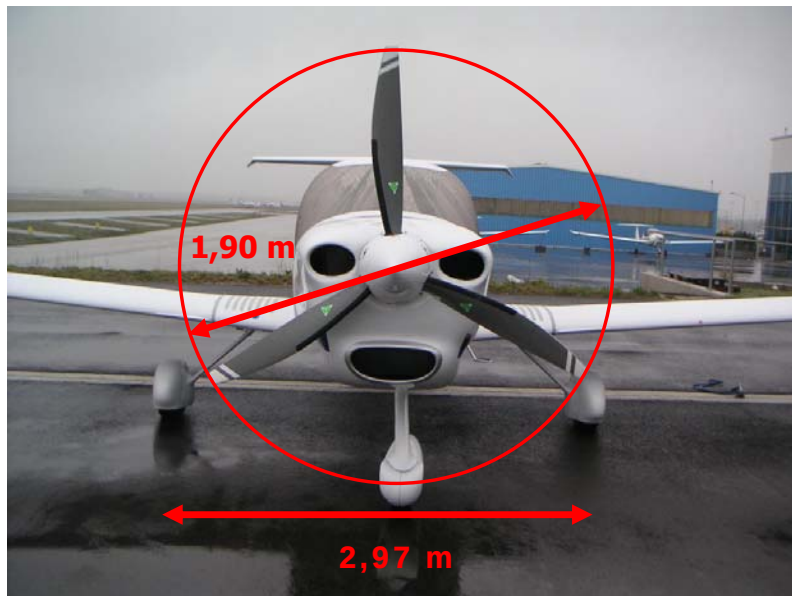
3

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Dimensions



DA40 NG

4

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

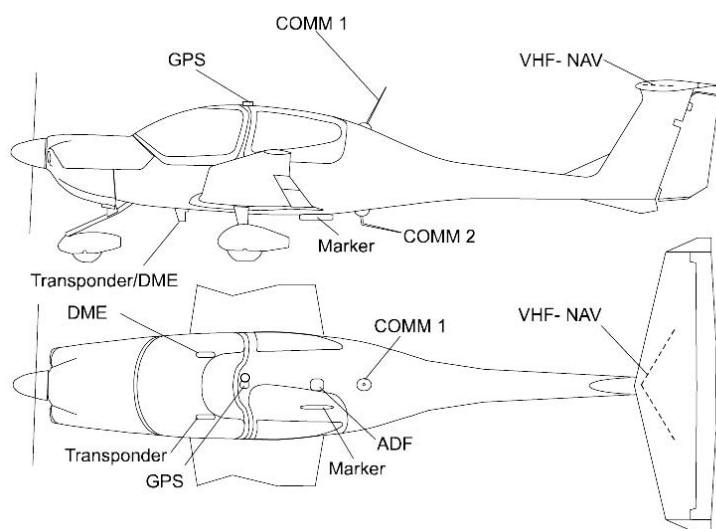
Diamond DA40 NG

Exterior



5

Antenna locations



DA40 NG

ELT and VHF COM 1 Antenna



DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

VHF COM 2 antenna



DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

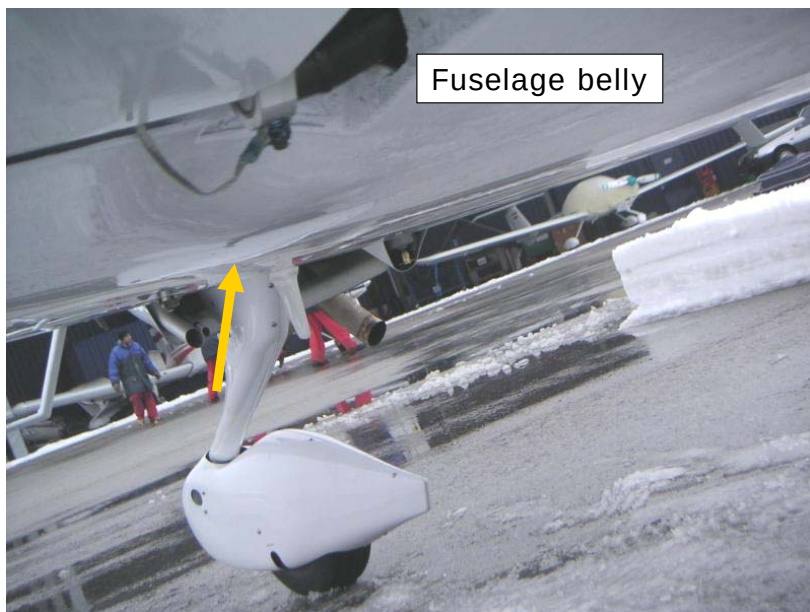
VHF NAV + GP Antennas



DA40 NG

9

Marker Antenna



DA40 NG

10

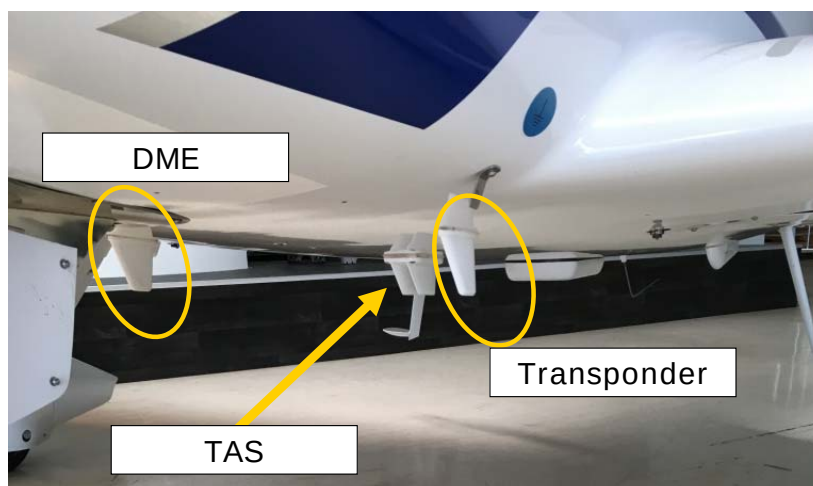
GPS antennas



DA40 NG

11

DME, TXPDR, TAS Antenna



DA40 NG

12

ADF antenna



DA40 NG

13

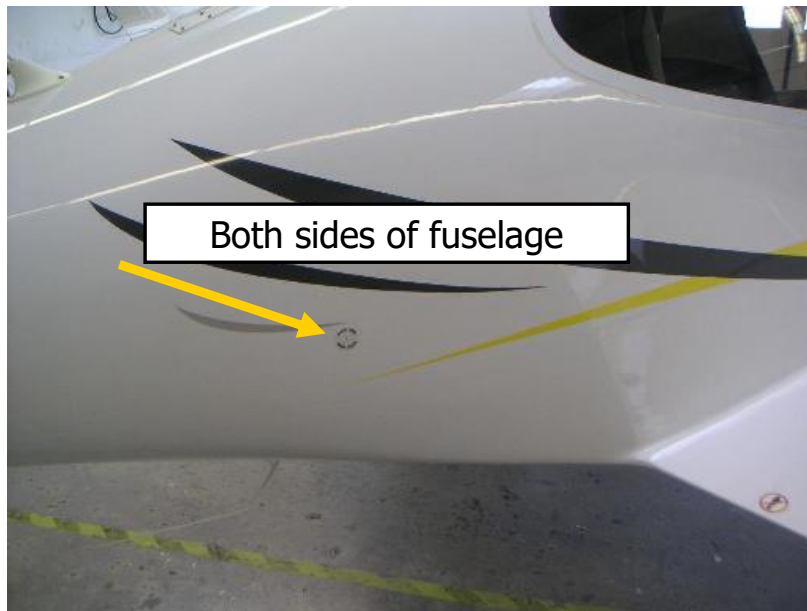
Pitot Probe



DA40 NG

14

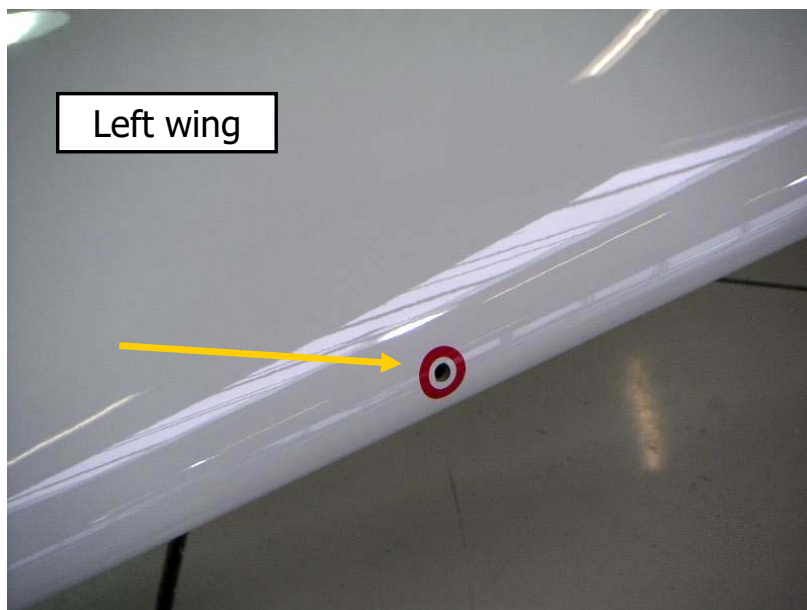
Static Ports



DA40 NG

15

Lift Detector (Stall Warning)



DA40 NG

16

Landing/Taxi Light



DA40 NG

17

Engine Oil and Gear Oil



DA40 NG

18

Engine Oil and Gear Oil



DA40 NG

19

Engine Oil and Gear Oil



DA40 NG

20

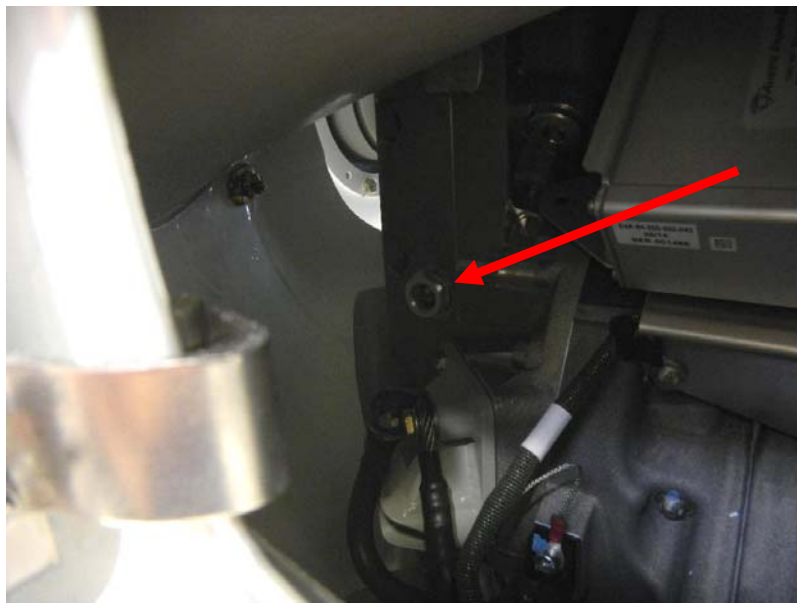
Engine Oil



DA40 NG

21

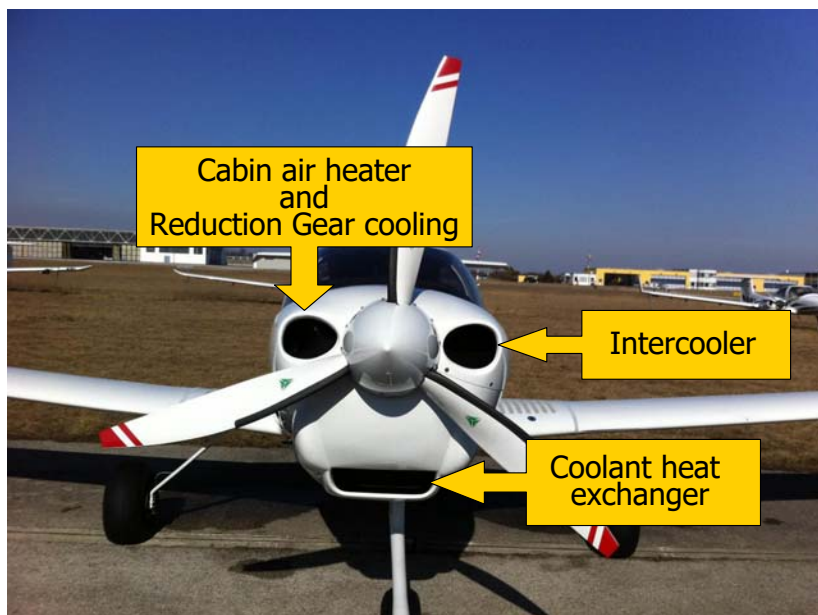
Gear Oil



DA40 NG

22

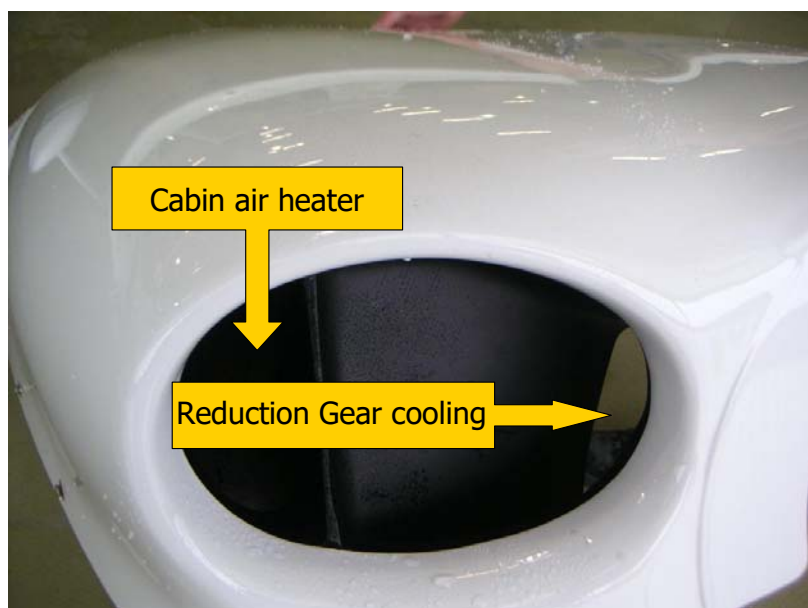
Air Inlets



DA40 NG

23

Air Inlets



DA40 NG

24

Coolant Radiator Inlet Baffle

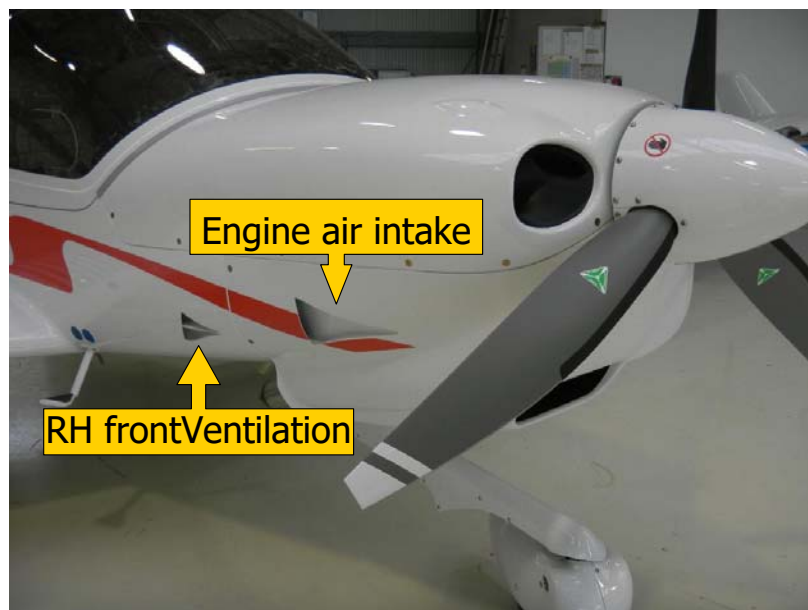


Must be installed when OAT at take-off is below -30° C, but operation with baffle only up to 0° C

DA40 NG

25

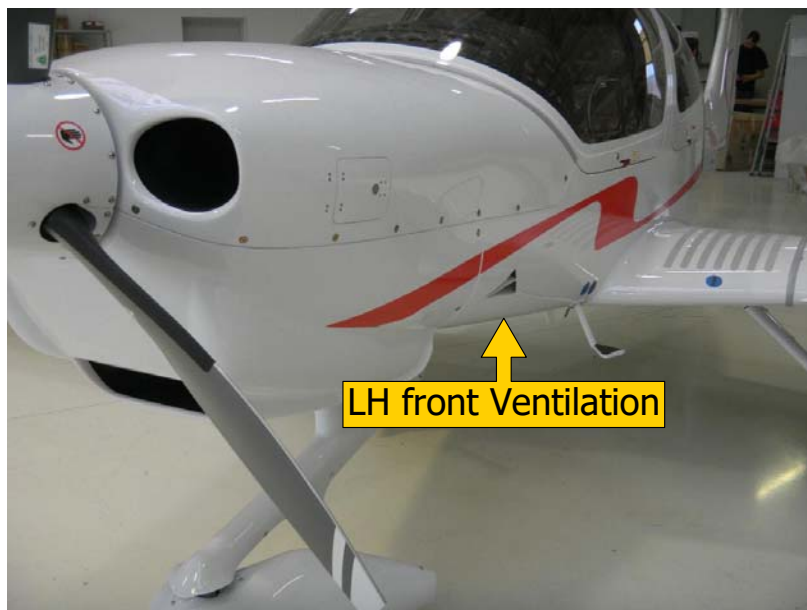
Air Inlets



DA40 NG

26

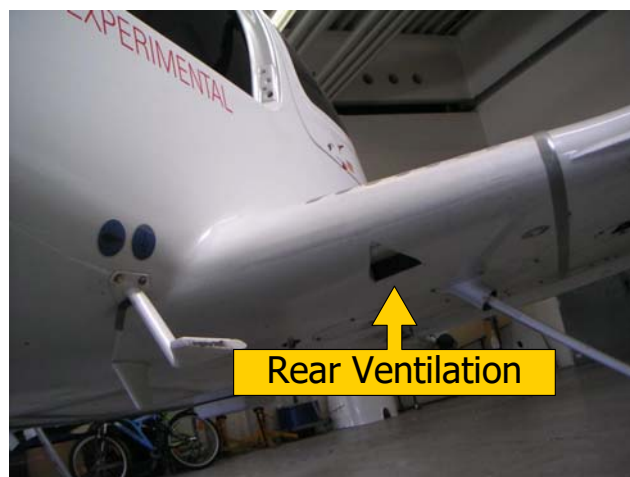
Air Inlets



DA40 NG

27

Air Inlets



May be closed in cold weather with „Ventilation inlet baffle“
With ventilation inlet baffle installed: max OAT for Take-off:
15° C

DA40 NG

28

Operating Limitations



29

Temperature Limitations

- **Airplane temperature:**
 - Below -20° C:
 - Pre-heating required for engine (until oil temp. above -20° C, between -15° C and -10° C recommended) and cockpit
 - Below -30° C:
 - Batteries must be pre-heated
 - Below -40° C:
 - Operation prohibited
- **Outside Air Temperature at TKOF:**
 - Below -30° C:
 - Coolant radiator inlet baffle must be installed
 - Above 0° C:
 - Coolant radiator inlet baffle must be removed
 - Above 15° C:
 - Ventilation inlet baffle must be removed
 - Above 20° C:
 - Fuel cooler inlet baffle must be removed

30

Altitude Limitation



- Maximum operating altitude:
16.400 ft (5.000 m) Pressure Altitude

DA40 NG

31

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Other Limitations



- Aerobatics, spinning, more than 60° bank are prohibited
- Canopy and passenger door must be unlocked during operation
- Max. demonstrated crosswind:
25 kts

DA40 NG

32

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Other Limitations

- No Take-off for Night-VFR or IFR with an empty main battery
- No engine start with external power because of an empty main battery if IFR or Night-VFR flight is planned. Battery must be charged first.
- No IFR flight if seal on the Emergency Switch is broken

Intentionally blank

Diamond DA40 NG



35

Mass

		Option „574“	Option „662“
Empty (typical)	900 kg		
Max TKOF	1280 kg		1310 kg
Max Ramp	+ 4 kg		
	1216 kg	1280 kg	
Max ZFW	1200 kg	1265 kg	
Min flight	940 kg		

36

Max Landing Mass

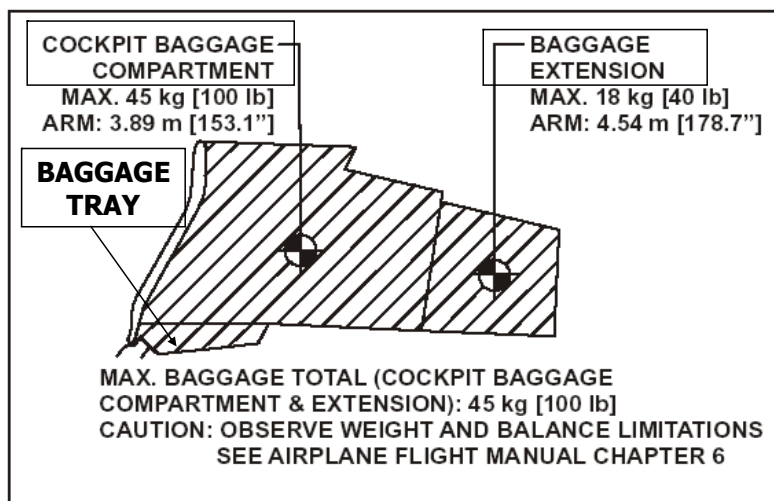
- If the Max Landing Mass is 1216 kg a landing with a mass between 1216 kg and 1280 kg is an „Abnormal Operating Procedure“
- Same as the „Normal Procedure“, observing higher speed: 77 KIAS

Max. Baggage Load

	Standard	Options	
		Baggage Extension	„Short“ Baggage Extension
Standard compartment	30 kg	45 kg	30 kg
Max in „tube“ (if installed)	5 kg	---	---
Baggage Extension	---	18 kg	15 kg
Total	35 kg	45 kg	45 kg

Mass

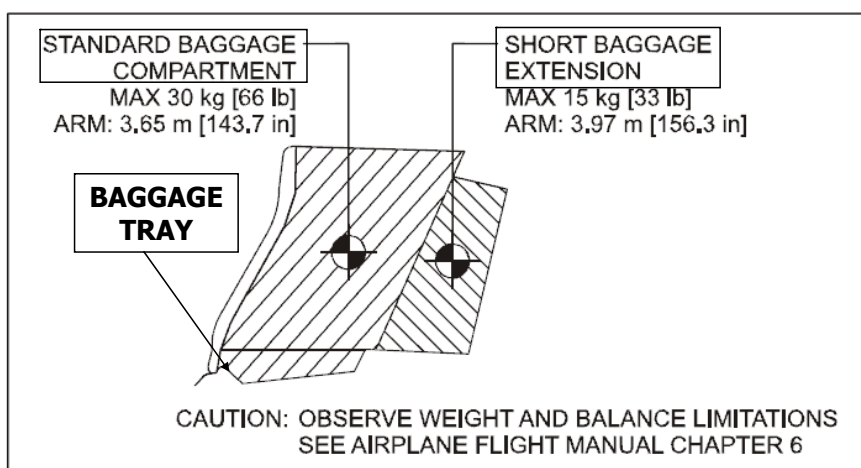
Option „Baggage Tray + Extended Compartment“



DA40 NG

39

„Short“ Baggage Extension



DA40 NG

40

Mass

Attention!
JET fuel and Diesel are heavier than AVGAS!

Typical fuel weight:

JET A1:	Diesel:
0,8 kg/ltr	0,84 kg/ltr
3,03 kg/USG	3,2 kg/USG

Intentionally blank

Diamond DA40 NG



43

Characteristic Speeds

V_{NO}	130 KIAS
V_{NE}	172 KIAS
V_O -1180 kg	101 KIAS
V_O 1081-1180 kg	108 KIAS
V_O 1181-1280 kg	113 KIAS

DA40 NG

44

Characteristic Speeds



V _{SO} (1310 kg)	60 KIAS
V _{S1} (1310 kg)	66 KIAS

V _{FE} (Flaps T/O)	110 KIAS
V _{FE} (Flaps LDG)	98 KIAS

DA40 NG

45

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Characteristic Speeds



KIAS at	940 kg	1000 kg	1100 kg	1200 kg	1280 kg +above
V _R Flaps T/O	56	58	61	65	67
V ₅₀ up to 50 ft Flaps T/O	62	65	67	70	72
V _Y up to safe altitude Flaps T/O	72				
V _{climb} climb Flaps UP	88				

DA40 NG

46

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Characteristic Speeds



KIAS at	940 kg	1000 kg	1100 kg	1200 kg	1216 kg	1280 kg +above
V _{ref} Approach Flaps LDG	66	68	72	76	76	77
V _{ref} Approach Flaps T/O	68	70	74	77	77	78
V _{ref} Approach Flaps UP	71	73	78	82	82	83
Minimum V _{Go Around} Flaps T/O	72					

DA40 NG

47

Characteristic Speeds



Best gliding Flaps UP, prop windmilling	88 KIAS
1:9,7	1,59 NM / 1000 ft
Without wheel fairings:	
1:9,4	1,54 NM / 1000 ft

DA40 NG

48

Stalling Speeds

1000 kg



55

58

V_{so}

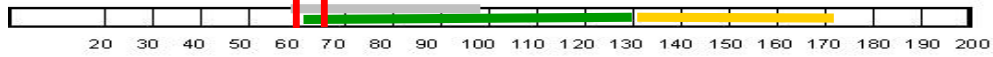
V_{s1}

60

66



1310 kg



DA40 NG

49

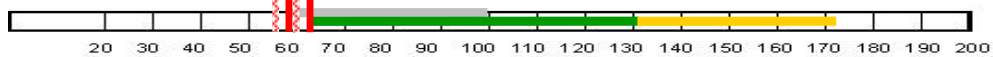
DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Stalling Speeds

1000 kg



56

59

V_{so}

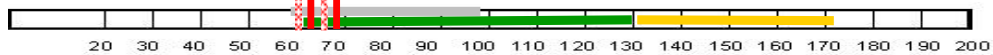
V_{s1}

63

68



1310 kg



DA40 NG

50

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Stalling Speeds

1000 kg



61

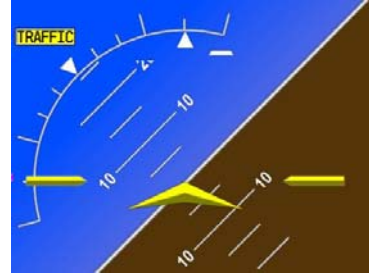
64

V_{so}

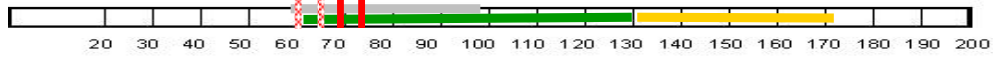
V_{s1}

69

74



1310 kg



DA40 NG

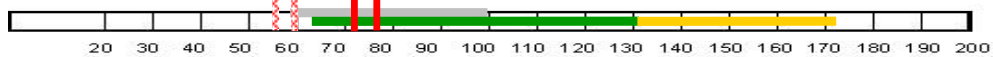
51

© Diamond Flight Training

Compiled by Peter Schmidleitner

Stalling Speeds

1000 kg



72

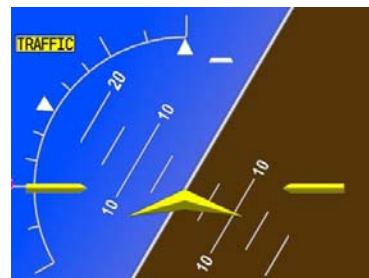
76

V_{so}

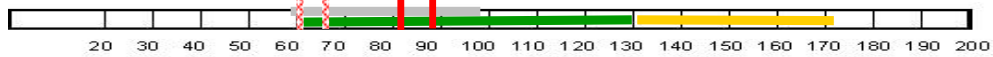
V_{s1}

82

88



1310 kg



DA40 NG

52

© Diamond Flight Training

Compiled by Peter Schmidleitner

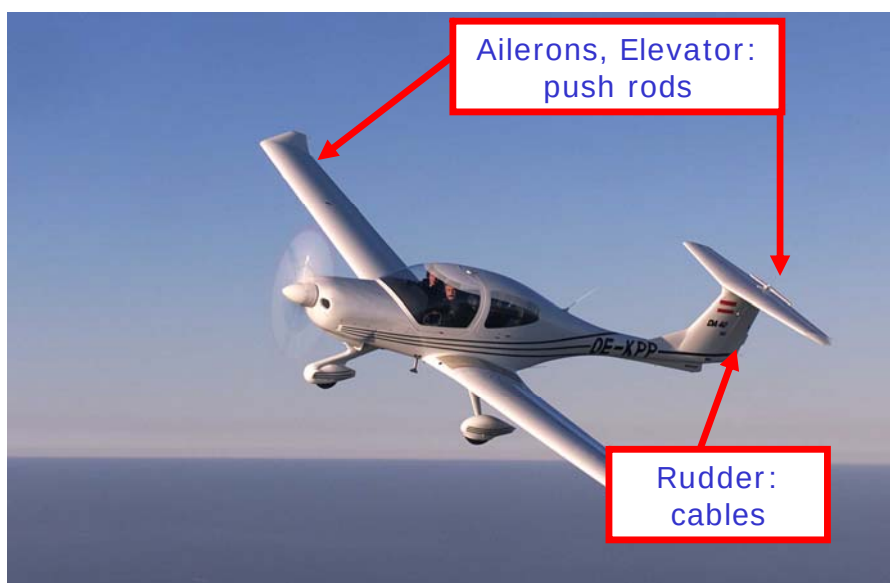
Diamond DA40 NG



Flight Controls

53

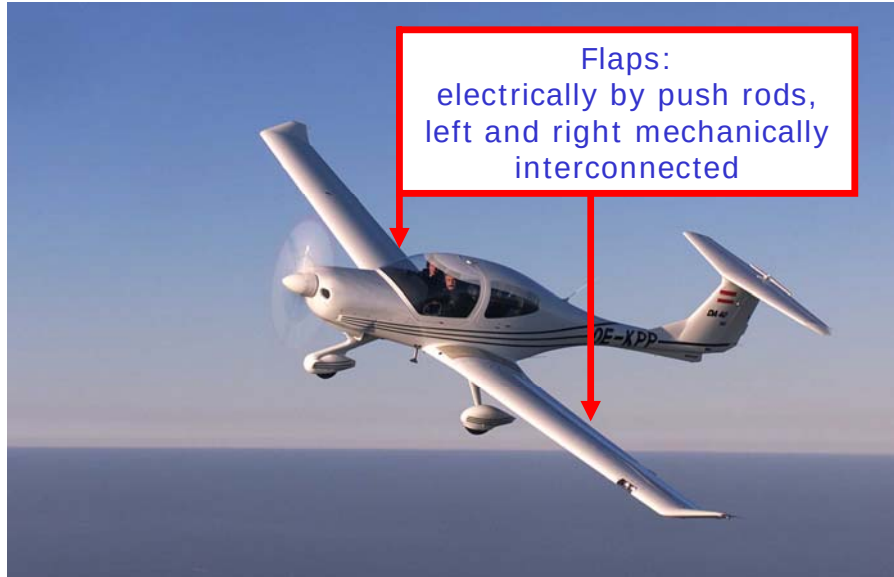
Flight Control Operation



DA40 NG

54

Flight Control Operation



DA40 NG

55

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Intentionally blank

DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG

Instrument Panel



57

Instrument Panel

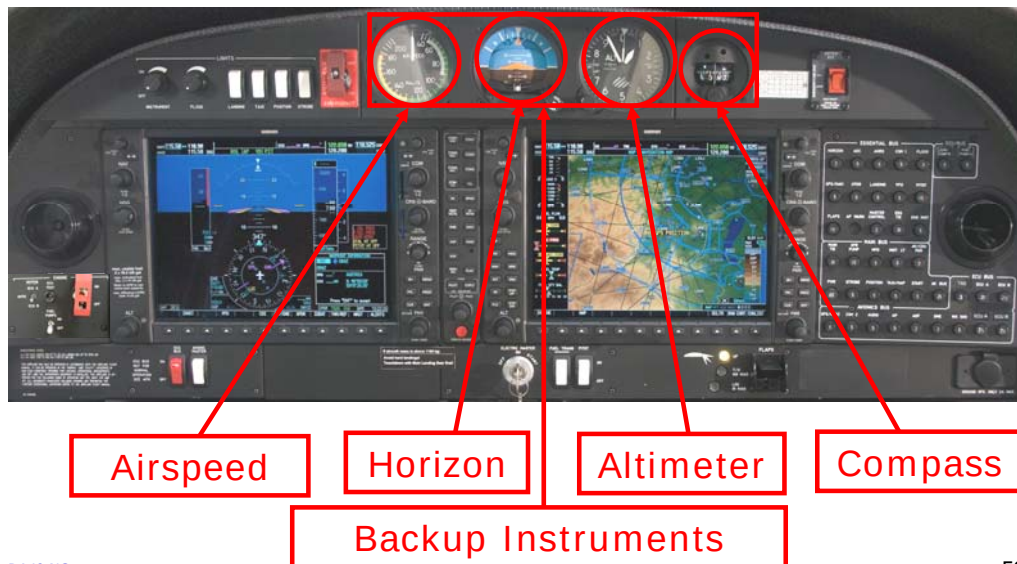


Garmin 1000

DA40 NG

58

Instrument Panel



DA40 NG

59

Instrument Panel



DA40 NG

IFR flights are not permitted when the seal on the emergency switch is broken.60

Instrument Panel



Lights

ELT

DA40 NG

61

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Instrument Panel



Circuit breakers

DA40 NG

62

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Instrument Panel



Essential Bus

Electric Master
Starter

DA40 NG

63

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Instrument Panel



Engine control

DA40 NG

64

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Instrument Panel



Avionic
Master

Fuel x-fer pump,
Pitot heat

DA40 NG

65

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Instrument Panel



Flaps

DA40 NG

66

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Alternate Static Valve



DA40 NG

67

Alternate Static Valve



DA40 NG

68

Diamond DA40 NG



69

Garmin 1000

- For a detailed description of the G1000 integrated instrument system consult the instruction modules „DA42 G1000, parts 1 to 3“

70

Garmin 1000



PFD

Audio Panel

MFD

Primary Flight Display

Multi Function Display

DA40 NG

71

DIAMOND FLIGHT TRAINING

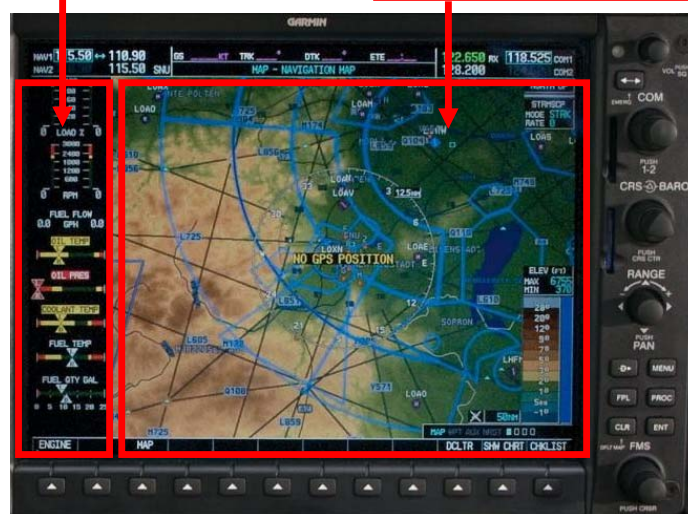
© Diamond Flight Training

Compiled by Peter Schmidleitner

MFD

EIS / Engine Indication System

Multi Function Display



DA40 NG

72

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

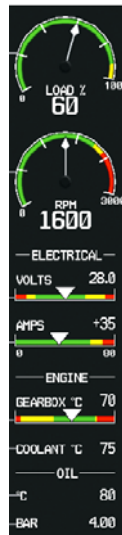
Compiled by Peter Schmidleitner

Engine Indication System

Default page



Display when pushing the SYSTEM softkey



Display when pushing the FUEL softkey



DA40 NG

© Diamond Flight Training

73

DIAMOND FLIGHT TRAINING

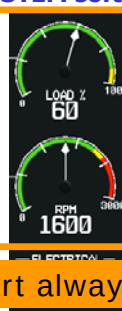
Compiled by Peter Schmidleitner

Engine Indication System

Default page



Display when pushing the SYSTEM softkey



Display when pushing the FUEL softkey



Upper part always the same

LOAD is not measured, but calculated from common rail fuel pressure, ECU commanded injection time, manifold pressure and Prop RPM.

Fuel Flow is not measured, but calculated from fuel pressure and ECU commanded injection time.

Readouts and labels and pointers appear white to indicate normal operation yellow or red to indicate caution or warning conditions

DA40 NG

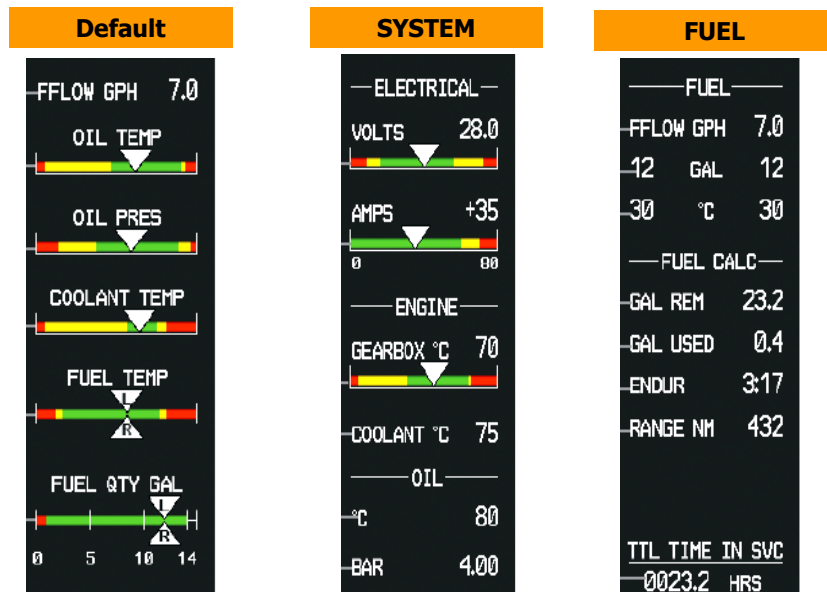
© Diamond Flight Training

74

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Engine Indication System



DA40 NG

© Diamond Flight Training

75

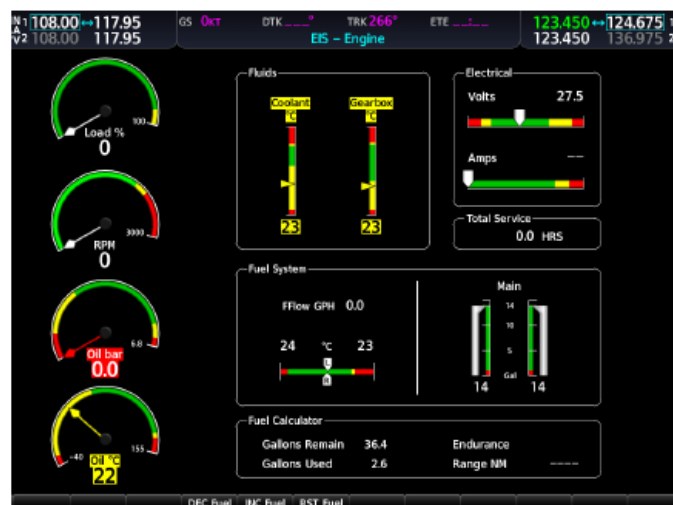
DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Engine Indication System with G1000 NXi



Display when pushing the ENGINE button:



DA40 NG

© Diamond Flight Training

75

DIAMOND FLIGHT TRAINING

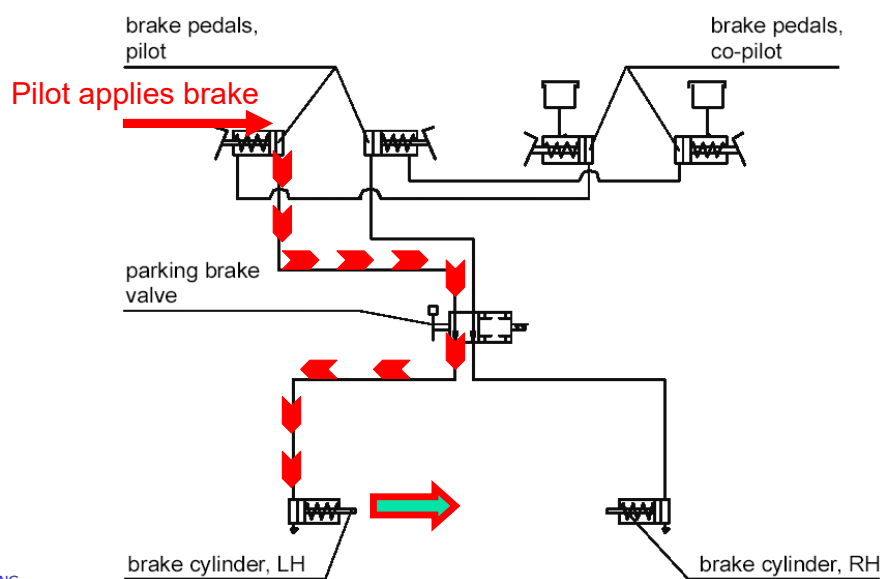
Compiled by Peter Schmidleitner

Diamond DA40 NG



77

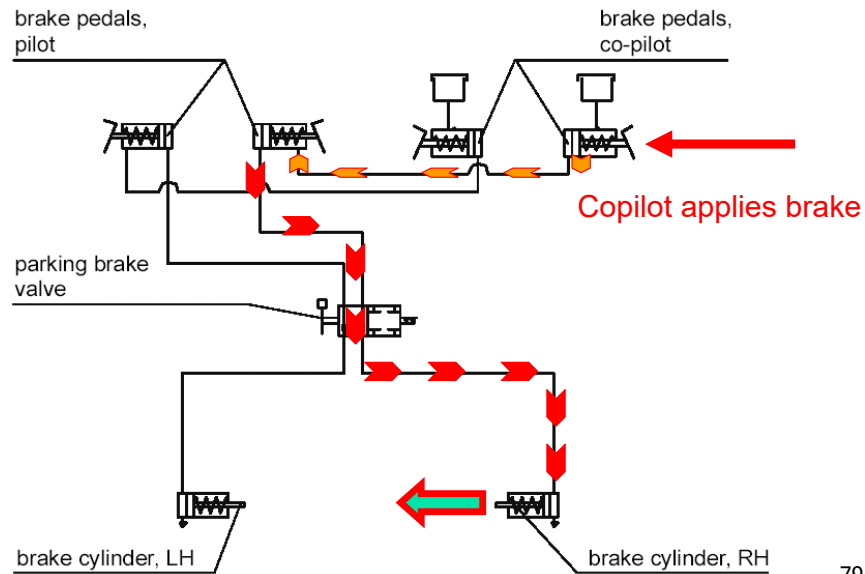
Hydraulic brakes



DA40 NG

78

Hydraulic brakes



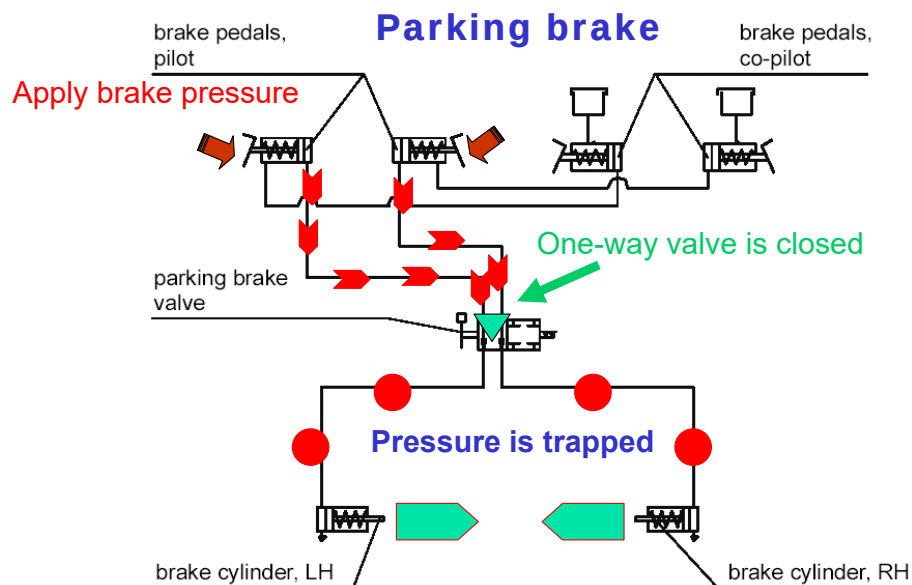
DA40 NG

79

© Diamond Flight Training

Compiled by Peter Schmidleitner

Hydraulic brakes



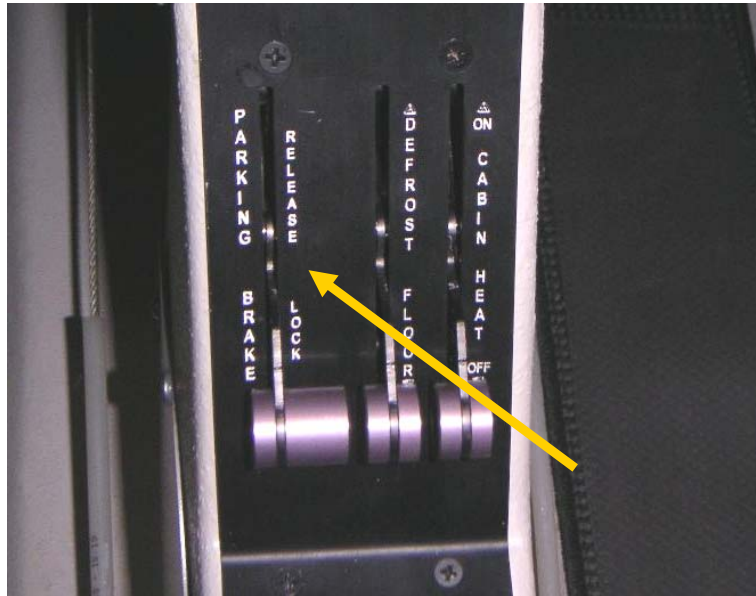
DA40 NG

80

© Diamond Flight Training

Compiled by Peter Schmidleitner

Parking brake



DA40 NG

81

Intentionally blank

DA40 NG

Diamond DA40 NG



83

Power plant

- 1 Austro Engine E4-A
- Four cylinders, liquid-cooled
 - 1991 ccm
- Common-rail direct injection
- Reduction gear 1:1,69
- Dual digital engine control
- Turbocharger
- Torsion vibration damper isolates engine from propeller
- Max. power: **100 %** (5 minutes time limit)
123,5 kW (165,6 DIN-HP) at 2300 RPM
- Max cont. power: **92 %**
113,6 kW (152,3 DIN-HP) at 2100 RPM

DA40 NG

84

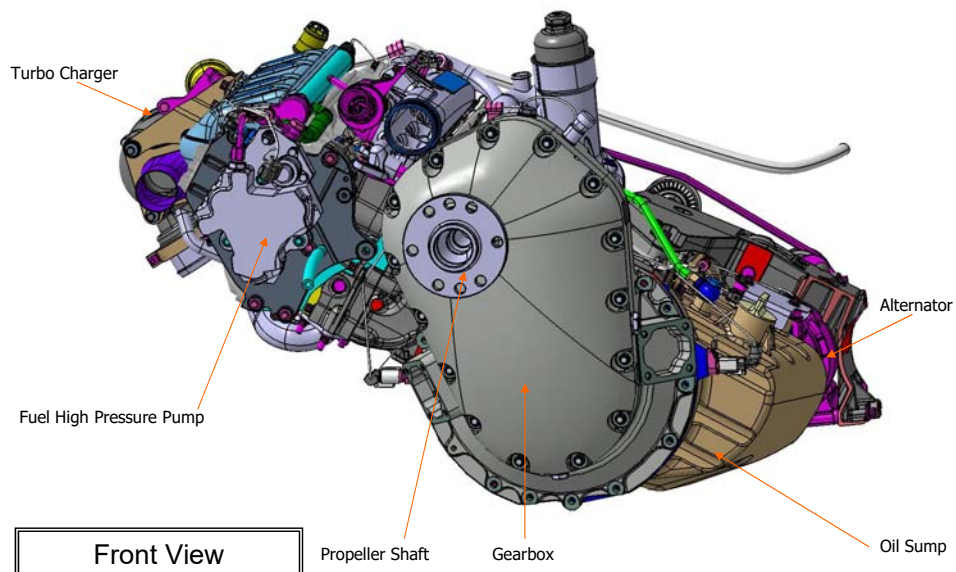
Austro Engine E4-A



DA40 NG

85

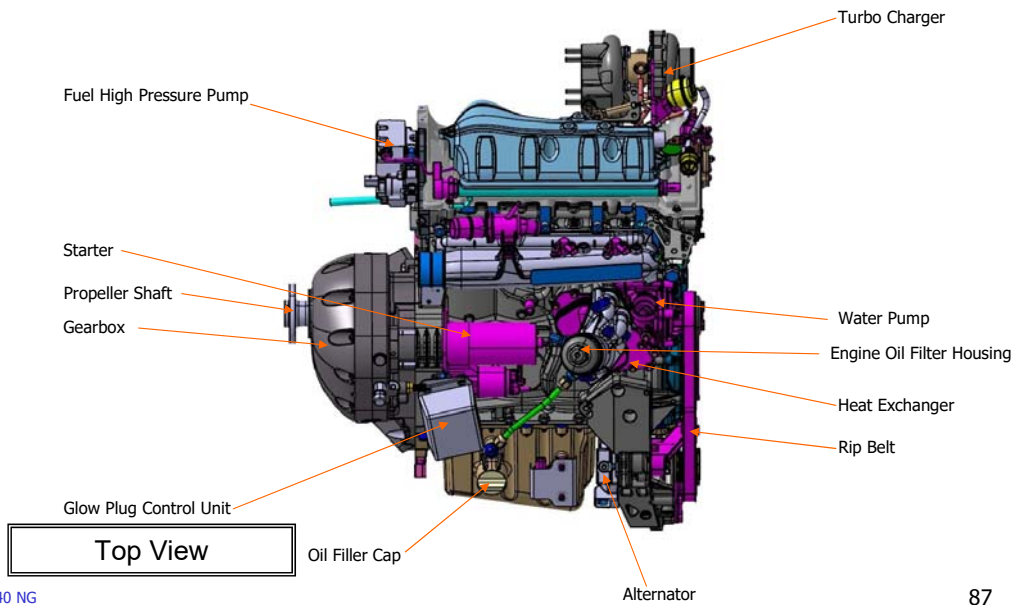
Austro Engine E4-A



DA40 NG

86

Austro Engine E4-A



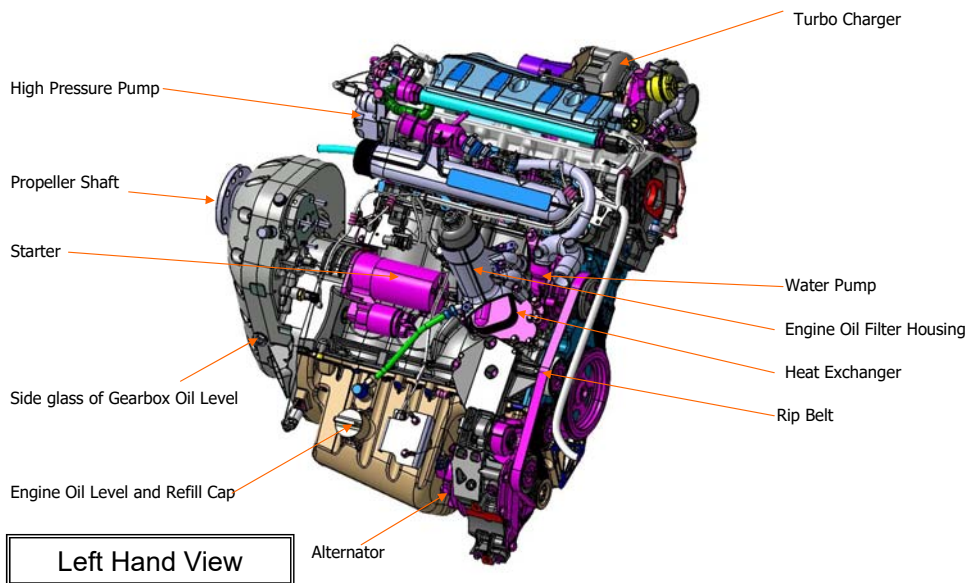
DA40 NG

87

© Diamond Flight Training

Compiled by Peter Schmidleitner

Austro Engine E4-A



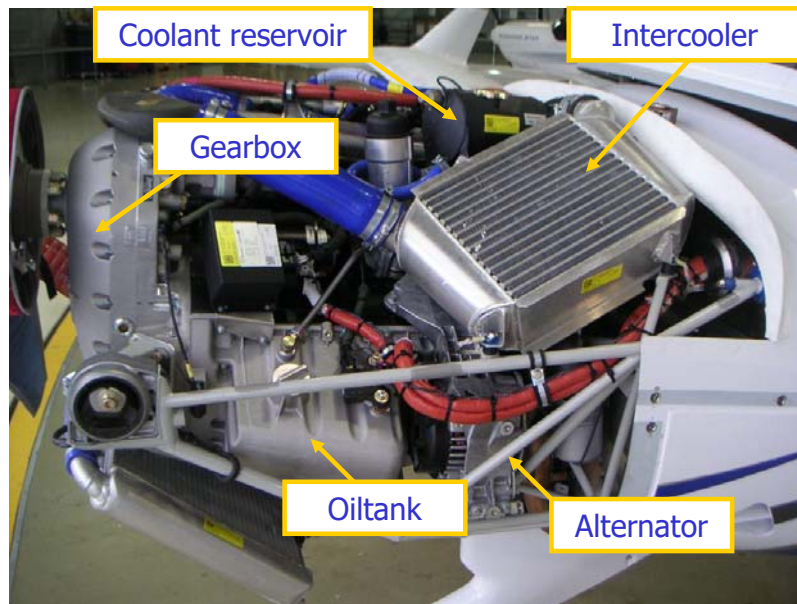
DA40 NG

88

© Diamond Flight Training

Compiled by Peter Schmidleitner

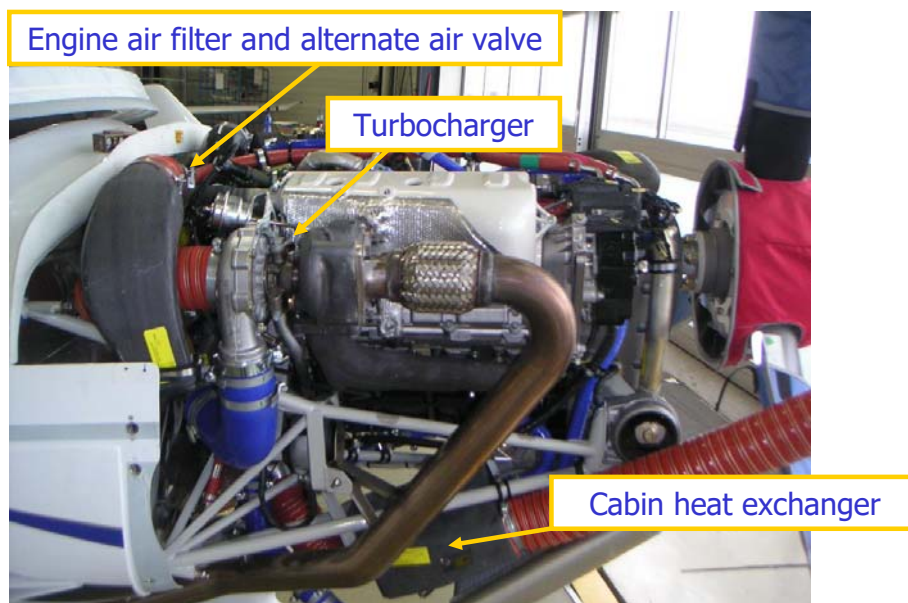
Austro Engine E4-A



DA40 NG

89

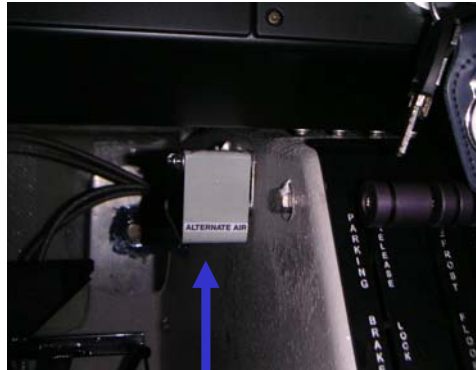
Austro Engine E4-A



DA40 NG

90

Alternate air



Alternate air lever



Alternate air valve

DA40 NG

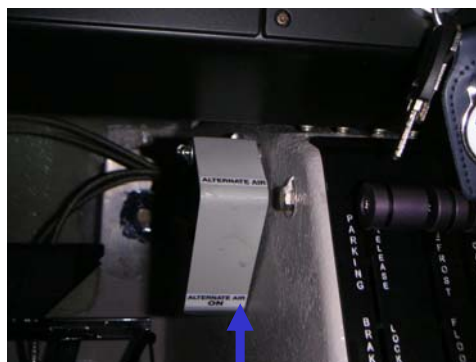
91

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Alternate air



Alternate air lever pulled



Alternate air valve open

DA40 NG

92

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Use of Alternate Air



ENGINE TROUBLESHOOTING

Alternate air OPEN

ENGINE RESTART

Alternate airAS REQUIRED

UNINTENTIONAL FLIGHT INTO ICING

Alternate air OPEN as required

DA40 NG

93

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Austro Engine E4-A



DA40 NG

94

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

ECU – Engine Control Unit

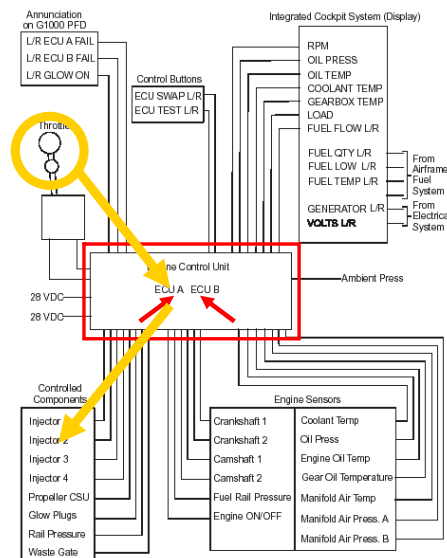
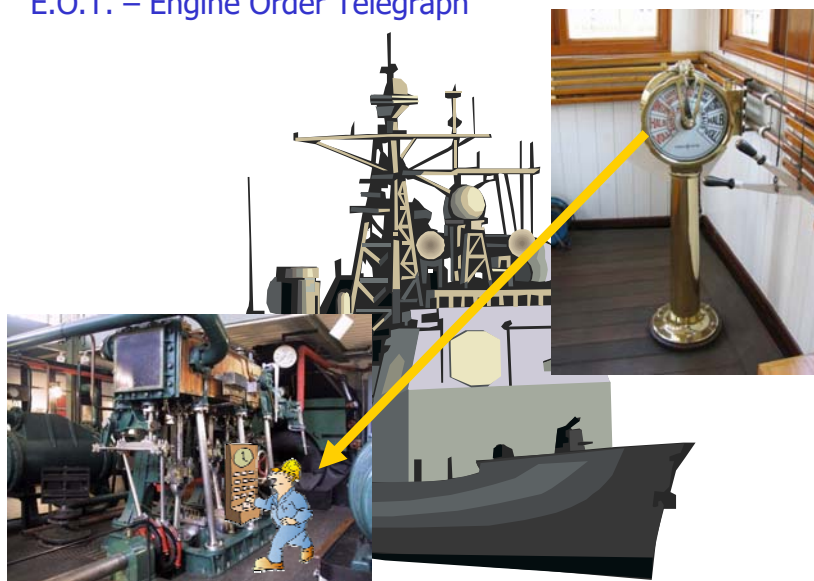


Figure 1: An Engine Control System Schematic Diagram

DA40 NG

95

E.O.T. – Engine Order Telegraph



DA40 NG

96

Engine Control Unit

- ECU „VOTER switch“
- Normally in AUTO position
- Working ECU is automatically selected according operating hours or in case of malfunction

DA40 NG

97

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

ECU test button

Test on ground (PWR lever idle)



DA40 NG

98

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

ECU test

BEFORE TAKE OFF CHECK

11	Power lever	IDLE	11
12	MFD	EIS – SYSTEM	12
13	Engine instruments	CHECKED	13

*Engine temperatures must be in green range before performing ECU test.
(For gearbox min.38° recommended). For warm up max power 50%.*

14	VOTER switch	A, AUTO, B, AUTO	14
----	--------------------	------------------	----

ECU TEST

ECU test button.....press and hold

"ECU A/B fail" ON

Prop cycling 2 times to 1800 RPM

"ECU A/B fail" OFF

ECU test button..... release

15	ECU test	PERFORMED	15
----	----------------	-----------	----

DA40 NG

99

ECU FAIL Caution

- An ECU FAIL caution is caused by various types of malfunctions. These include internal ECU problems, sensor failure or insufficient performance of air-, fuel- or electrical supply system (e.g. air filter icing).
- Depending on the type of failure, the ECU failure cautions are either
 - 'non latched', i.e. the caution message disappears after the cause of the caution is no longer present or
 - 'latched', i.e. the caution message remains until cleared through maintenance action, and Austro Engine has to be informed.
- Clearing of 'non-latched' caution:
 - on the active ECU: clears by itself
 - on the passive ECU: by switching to that ECU with the voter switch.

DA40 NG

100

ECU FAIL Caution

ECU A OR B FAIL

ON GROUND

- | | | | |
|---|-------------------------|--------------|---|
| 1 | Alternate Air | check CLOSED | 1 |
| 2 | Fuel pump | OFF | 2 |
| 3 | VOTER switch | check AUTO | 3 |
| 4 | Other ECU caution | check OFF | 4 |

Clearing procedure:

- | | | | |
|---|--------------------|-------------------|---|
| 5 | VOTER switch | set to failed ECU | 5 |
| | | Wait 5 seconds | |
| 6 | Voter switch | AUTO | 6 |
- If ECU caution persists terminate flight preparation

DA40 NG

101

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

ECU FAIL Caution

ECU A OR B FAIL

DURING FLIGHT

Remark: in case of ECU fail the system automatically switches to the other ECU

- | | | | |
|---|------------------------|--------------------------|---|
| 1 | Alternate Air | OPEN | 1 |
| 2 | Fuel pump | ON | 2 |
| 3 | Circuit breakers | CHECK/RESET if necessary | 3 |
| 4 | VOTER switch | check AUTO | 4 |
- If ECU caution persists:
⇒ Land at nearest suitable airfield
 - If additional engine problems are observed:
⇒ Go to **Emergency Checklist page 4**
ENGINE TROUBLESHOOTING

Remark: after landing the clearing procedure for "ECU FAIL ON GROUND" may be used.

ECU A AND B FAIL SIMULTANEOUSLY

DURING FLIGHT

- Go to **Emergency Ckl page 4** ENGINE TROUBLESHOOTING

DA40 NG

102

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Engine troubleshooting

→ If **ECU A AND B FAIL** simultaneously and ALL of the following conditions exist:

- indicated **LOAD** unchanged
- perceived **thrust is reduced**
- engine **noise level changes** or engine running rough

3 POWER leverIDLE for 1 second 3

4 POWER leverslowly increase to 1975 RPM 4

- If engine shows power loss during the POWER lever increase

5 POWER lever idle for 1 second 5

6 POWER lever slowly increase 6

stop prior to the RPM where former engine power loss was observed

Do not increase the POWER lever past the propeller speed of 1975 RPM or the setting determined in step 4. An increase of engine power beyond this setting leads into another power loss.

With this power setting the engine can provide up to 65% at the maximum propeller speed of 1975 RPM

7 Land at nearest suitable airfield..... 7

End of Checklist

DA40 NG

103

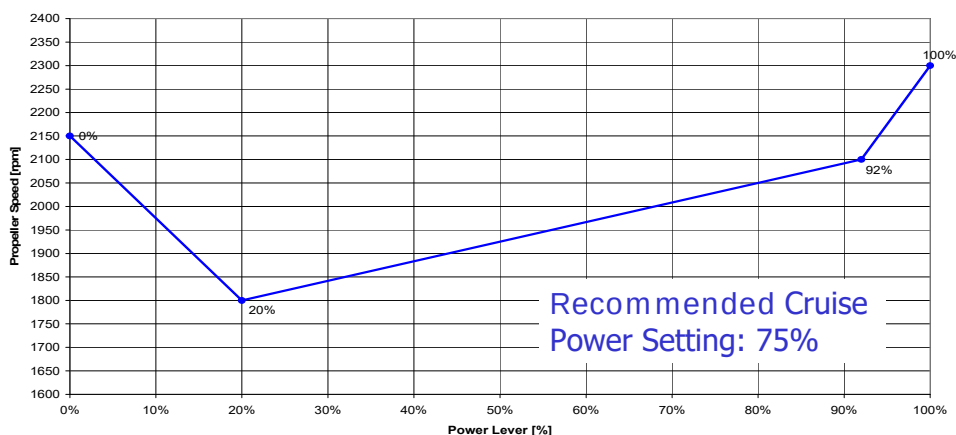
DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant

- Power lever selects „LOAD“ in %
- RPM automatically determined by selected power



DA40 NG

104

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Propeller

- 3-blade wooden propeller
- Constant speed
- Prop pitch set by ECU via an electro-mechanical actuator on the governor
- Governor operated by gearbox oil
 - Oil pressure up = pitch up = RPM down
 - Oil pressure down = pitch down = RPM up

DA40 NG

105

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

RPM malfunctions

OSCILLATING RPM

- | | | | |
|---|-----------------------------------|----------------------|---|
| 1 | Power lever | CHANGE SETTING | 1 |
| | ● If no success: | | |
| 2 | VOTER switch | SWAP between A and B | 2 |
| | ● If no success: | | |
| 3 | VOTER switch | AUTO | 3 |
| | Land at nearest suitable airfield | | |

RPM UNDERSPEED

- | | | | |
|---|-----------------------------------|----------------------|---|
| 1 | Power lever | AS REQUIRED | 1 |
| 2 | VOTER switch | SWAP between A and B | 2 |
| | ● If no success: | | |
| 3 | VOTER switch | AUTO | 3 |
| 4 | Power lever | AS REQUIRED | 4 |
| | Land at nearest suitable airfield | | |

DA40 NG

106

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

RPM malfunctions

RPM OVERSPEED

- | | | | |
|-----------------------------------|--------------------|-------------------------|----|
| 1 | Power lever | ADJUST to max. 2300 RPM | 1 |
| 2 | Airspeed..... | 88 KIAS | 2 |
| 3 | Flaps | UP | 3 |
| ✦ If RPM stabilized below 2300: | | | |
| 4 | Airspeed..... | AS REQUIRED | 4 |
| 5 | Power lever | AS REQUIRED | 5 |
| but do not exceed 2300 RPM | | | |
| ✦ If RPM still above 2300: | | | |
| 6 | VOTER switch | SWAP between A and B | 6 |
| ● If no success: | | | |
| 7 | VOTER switch | AUTO | 7 |
| adjust RPM with power lever | | | |
| Land at nearest suitable airfield | | | |
| If increased climb rate required: | | | |
| 8 | Flaps | T/O | 8 |
| 9 | Airspeed..... | 72 KIAS | 9 |
| 10 | Power lever | ADJUST to max. 2300 RPM | 10 |

DA40 NG

107

Fuel pumps

- 1 engine driven pump
- this high pressure pump feeds the common rail
- (additional electrical fuel pumps are part of the fuel system)

DA40 NG

108

Power plant limitations



- Max overspeed: 2500 RPM, max 20 sec.
- Oil pressure:
 - At idle: min 0,9 bar
 - At max continuous PWR: min 2,5 bar
 - Max: 6,5 bar
 - Normal: 2,5 – 6 bar
- Oil quantity: 5.0 – 7.0 liters
 - Max. oil consumption: 0.1 liters/hr
- Oil temperature: -30° C – 140 ° C
 - Normal: 50° C – 125° C

DA40 NG

109

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant limitations



- Gearbox temperature:
 - Min: -30° C
 - Min at full load: 35° C
 - Max: 120 ° C

The yellow cautionary range is for information only. There is no time limit associated with the cautionary temperature range.



DA40 NG

110

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant limitations



- Coolant temperature:
 - min -30° C for start up
 - min 60° C full load
 - max 105 ° C
- Fuel temperature:
 - min -25° C, max 60° C
- Fuel pressure:
 - min 4 bar, max 7 bar
 - no indication on G1000, but warning if below limit

DA40 NG

111

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant limitations



- Max. restart altitude:
 - 16.400ft PA for immediate restart
 - 15.000ft PA for immediate restart with engine software V33_1_05_19 or later
 - 10.000ft PA for restart within 2 minutes
- After more than 2 minutes cooldown a succesful restart may not be possible!
- Restart airspeed:
 - The propeller will be windmilling
 - Airspeed: 88 KIAS
 - If propeller has stopped: mechanical engine defect. Do not consider a starter assisted restart!

DA40 NG

112

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant limitations



- Intentional negative-g manoeuvres are not permitted

DA40 NG

113

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Starter limitations



- Operation: max 10 seconds
- 60 seconds cool down time

DA40 NG

114

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant fluid specifications

■ Jet Fuel:

- JET A, JET A-1 (ASTM D 1655)
- TS-1 (Russia, GOST 10227-86)
- TS-1 (Ukraine, GSTU 320.00149943.011-99)
- RT (Russia, GOST 10227-86)
- RT (Ukraine GSTU 320.00149943.007-97)
- No. 3 Jet Fuel (China, GB 6537-2006)
- JP-8 (F34) (USA, MIL-DTL-83133G-2010)

and blends thereof

*Minimum cetane number of 36
(EN ISA 5165/ASTM D613)
recommended*

- For additives see AFM!

■ Diesel Fuel

- No additives permitted!

DA40 NG

115

Power plant fluid specifications

■ Engine oil:

e.g. SHELL Helix Ultra
AEROSHELL Oil Diesel Ultra
AUSTRO ENGINE Aero

For more oil types see AFM

■ Gearbox oil:

Shell Spirax GSX 75W-80
Shell Spirax S6 GXME 75W-80

■ Coolant:

Distilled water + cooler protection
(BASF Glysantin Protect Plus/G48)

Mixture 50% / 50%: freezing point -38° C

For cold weather operation:

Mixture 40% / 60%: freezing point -53° C

DA40 NG

116

Engine operation



CHECK AFTER ENGINE START

16	Oil and Coolant temperature	CHECKED	16
17	Parking brake	RELEASED	17

After engine start max power 50% until
engine temperatures in green range

End of Checklist; see next page for "During taxi" – items

BEFORE TAKE OFF CHECK after line-up

Available power check (see pg.6) PERFORMED

Available Power Check:

10 sec. power MAX, RPM 2200 – 2300 (min. 2100 below -10°C), min. load acc. table below

Altitude [ft]	OAT								
	-35°C -31°F	-20°C -4°F	-10°C 14°F	0°C 32°F	10°C 50°F	20°C 68°F	30°C 86°F	40°C 104°F	50°C 122°F
0	94%						95%	92%	90%
2000							95%	92%	
4000							95%	92%	
6000				96%			95%	92%	
8000						95%	94%	91%	
10000				94%	93%	91%	88%		

DA40 NG

117

© Diamond Flight Training

Compiled by Peter Schmidleitner

Engine operation



PARKING CHECK

1	Parking brake	SET	1
2	Power lever	max 10% for 1 min.	2
3	ELT	CHECK not activated	3
4	Engine / System page	CHECKED	4
5	Engine / Fuel page	TTL TIME IN SVC NOTED	5
6	Avionic master	OFF	6
7	Electrical consumers except ACL (strobe)	OFF	7
8	Engine Master	OFF	8
9	ACL (strobe)	OFF	9

When engine indications x-out red:

10	Electric Master	OFF	10
11	Start key	REMOVED	11

End of Checklist

DA40 NG

118

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant warnings



OIL TEMP

OIL TEMPERATURE HIGH

- Check oil pressure
 - ❖ If too low:
 - ⇒ Reduce power
 - ⇒ Be prepared for loss of oil and engine fail; be prepared for emergency landing
 - ❖ If in green range:
 - ⇒ Reduce power
 - ⇒ Increase airspeed

OIL PRES

OIL PRESSURE LOW

- Reduce power
- Expect loss of oil
- Land at nearest suitable airfield
- Be prepared for engine fail

GBOX TEMP

GEARBOX TEMPERATURE HIGH

- Reduce power
- Increase airspeed
 - If gearbox temperature still in red range:
 - ⇒ Land at nearest suitable airfield
 - ⇒ Be prepared for engine fail

DA40 NG

119

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Power plant abnormalities



OIL pressure high

- ❖ On ground during warm up with low oil temperature
 - Reduce power until oil pressure green, continue warm up at reduced power
- ❖ During flight
 - Check oil temperature
 - Check coolant temperature
 - ❖ If temperatures within green range
 - ⇒ Oil pressure indication may be faulty; watch temperatures
 - ❖ If temperatures outside of green range
 - ⇒ Reduce power;
 - ⇒ Land at nearest suitable airfield, be prepared for engine fail

Oil pressure low

- Refer to Emergency Checklist page 6, "OIL PRES"

Oil temperature high

- Refer to Emergency Checklist page 6, "OIL TEMP"

Oil temperature low

- Increase power
- Reduce airspeed

DA40 NG

120

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

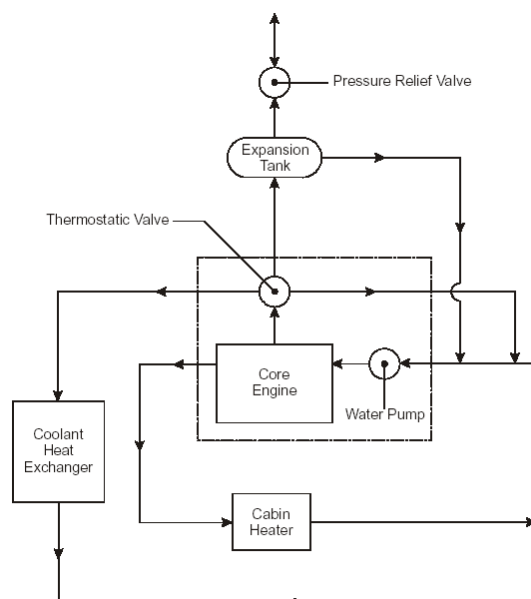
Compiled by Peter Schmidleitner

Diamond DA40 NG



121

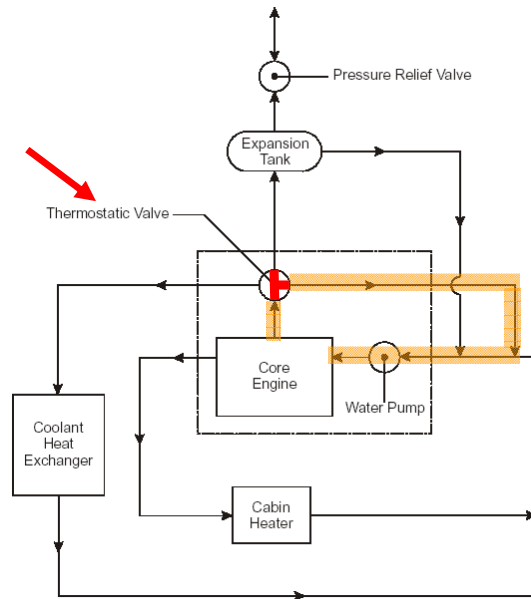
Cooling system



DA40 NG

122

Cooling system



Coolant Temp.
< ~80° C

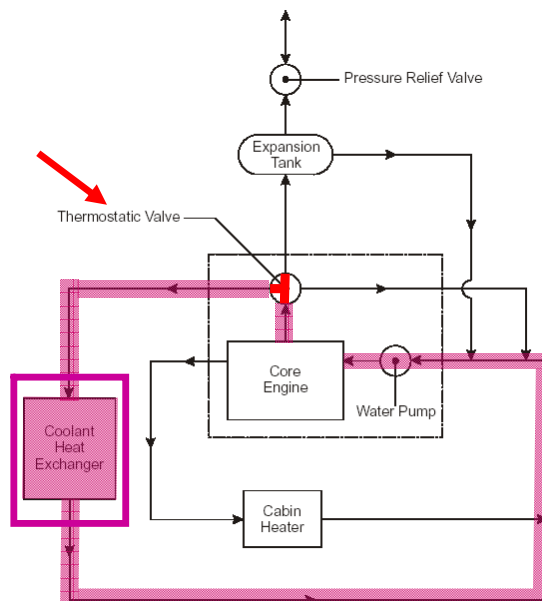
DA40 NG

123

© Diamond Flight Training

Compiled by Peter Schmidleitner

Cooling system



Coolant Temp.
> ~80° C

DA40 NG

124

© Diamond Flight Training

Compiled by Peter Schmidleitner

Cooling system

ENG TEMP

COOLANT TEMPERATURE HIGH

- Check "COOL LVL" caution message
 - ❖ If "COOL LVL" OUT:
 - ❖ During climb:
 - ⇒ Reduce power 10%
 - ⇒ Increase airspeed 10 KIAS
 - ⇒ If not returning to green range within 60 seconds: reduce power as far as possible and increase airspeed
 - ❖ During cruise:
 - ⇒ Reduce power
 - ⇒ Increase airspeed, if necessary descend
 - ⇒ Check coolant temperature in green range
 - If not returning to green range:
 - ⇒ land at nearest suitable airfield
 - ❖ If "COOL LVL" ON:
 - ⇒ Reduce power
 - ⇒ Expect loss of coolant fluid
 - ⇒ Be prepared for emergency landing

DA40 NG

125

Cooling system

Coolant temperature high

- Refer to **Emergency Checklist page 6**, "ENG TEMP"

Coolant temperature low

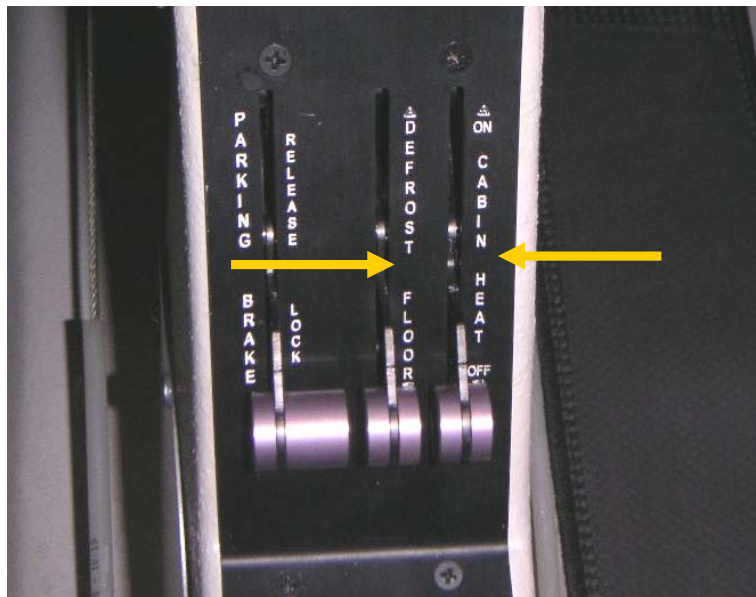
Remark: During low power descent from high altitude coolant temperature may decrease

- Check "COOL LVL" caution light
 - **If ON**
 - ⇒ Reduce power
 - ⇒ Expect loss of coolant fluid
 - ⇒ Be prepared for engine failure

DA40 NG

126

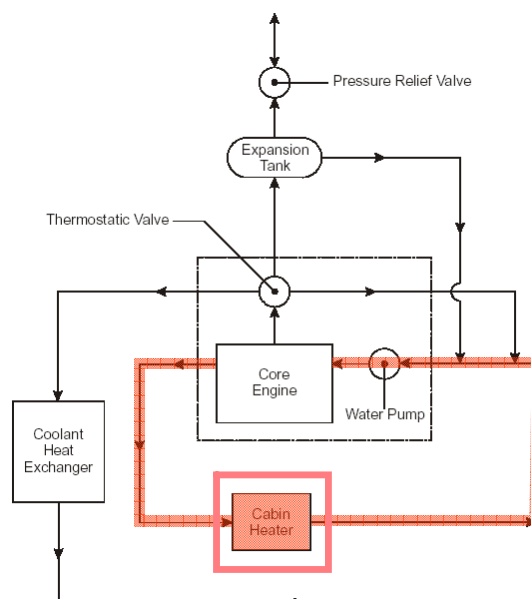
Cabin heat



DA40 NG

127

Cabin heat



Cabin Heat

DA40 NG

128

Ventilation

Cockpit



Front ventilation



DA40 NG

129

Ventilation

Cabin

Rear ventilation



DA40 NG

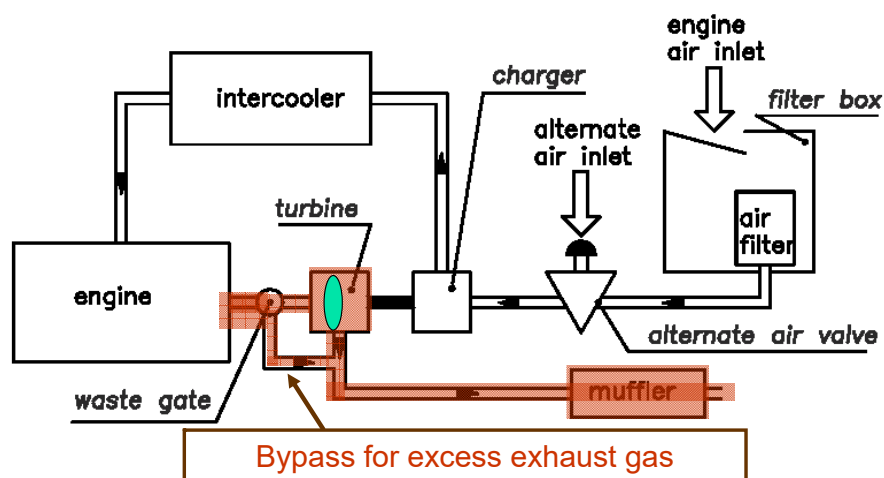
130

Diamond DA40 NG



131

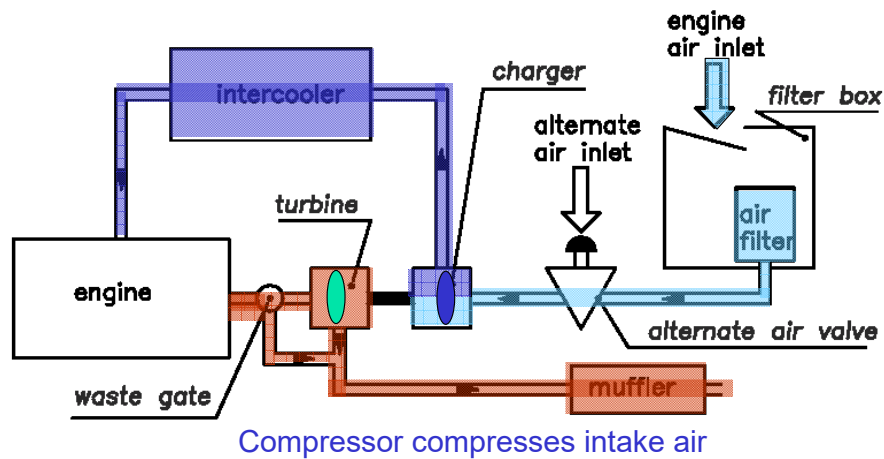
Turbo Charger



DA40 NG

132

Turbo Charger



DA40 NG

133

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Intentionally blank

DA40 NG

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner



135

Fuel System

- Standard Tanks
 - Capacity: 2 x 15 USG
 - 2 x 14 USG usable
 - = 28 USG usable
 - = 85 kg
 - At 75% power: ~ 6,5 USG/hr

136

Fuel System

- Long Range Tanks:
 - Capacity: 2 x 20,5 USG
 - 2 x 19,5 USG usable
 - = 39 USG
 - = 120 kg
- Max indicated fuel per tank: 14 USG
- Max. unbalance: 9 USG

DA40 NG

137

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel System



DA40 NG

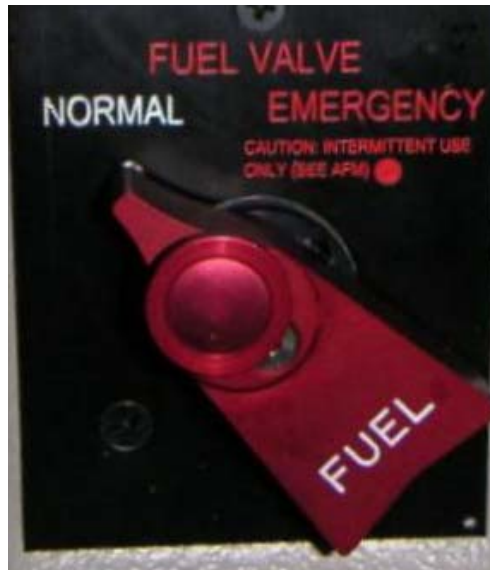
138

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

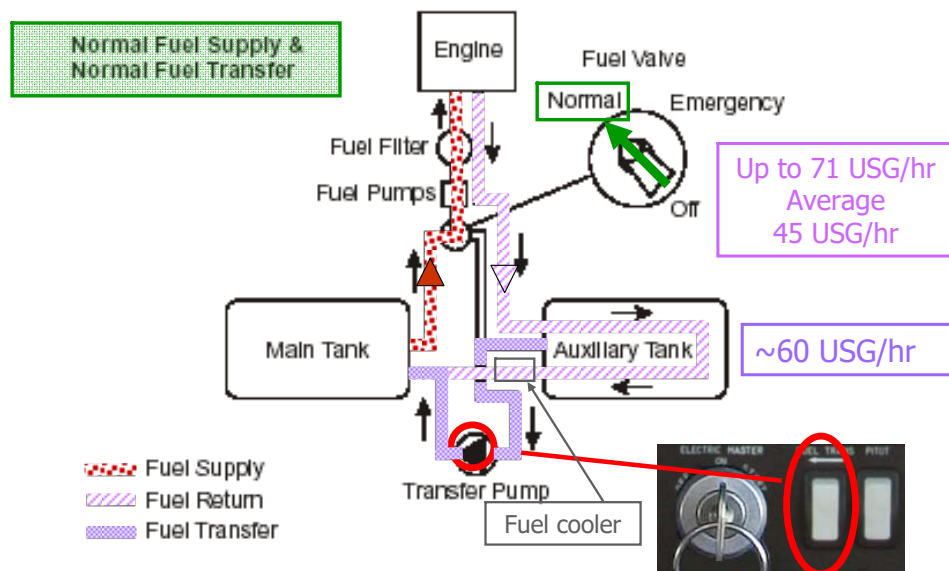
Fuel System



DA40 NG

139

Fuel System



DA40 NG

140

Fuel System

- The transfer pump is switched OFF automatically when
 - the auxiliary tank is empty
 - the main tank is full (14 USG)

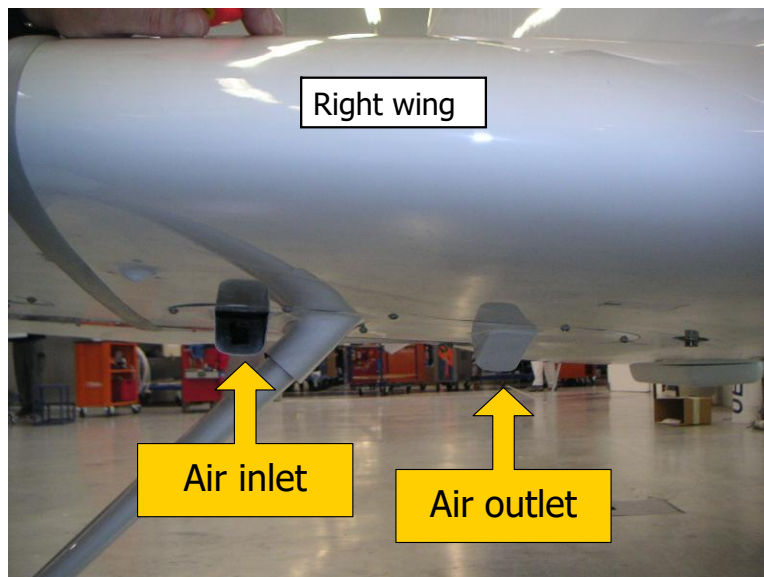
DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel cooler



DA40 NG

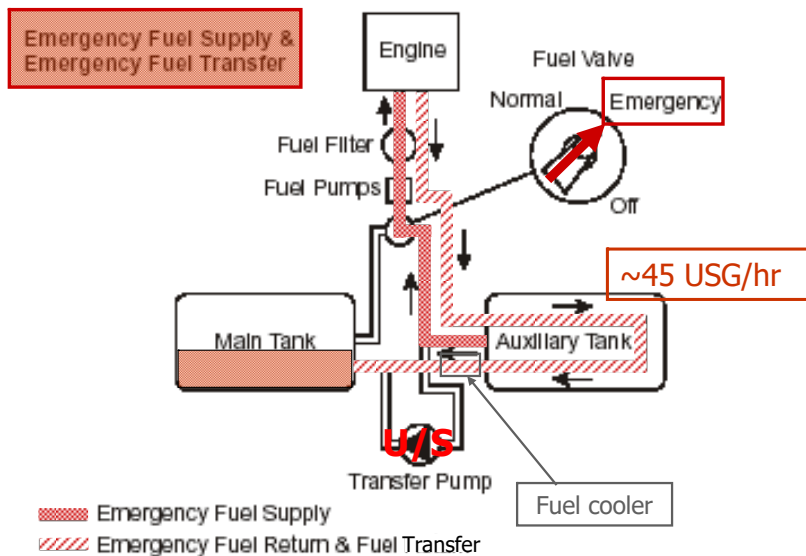
DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

142

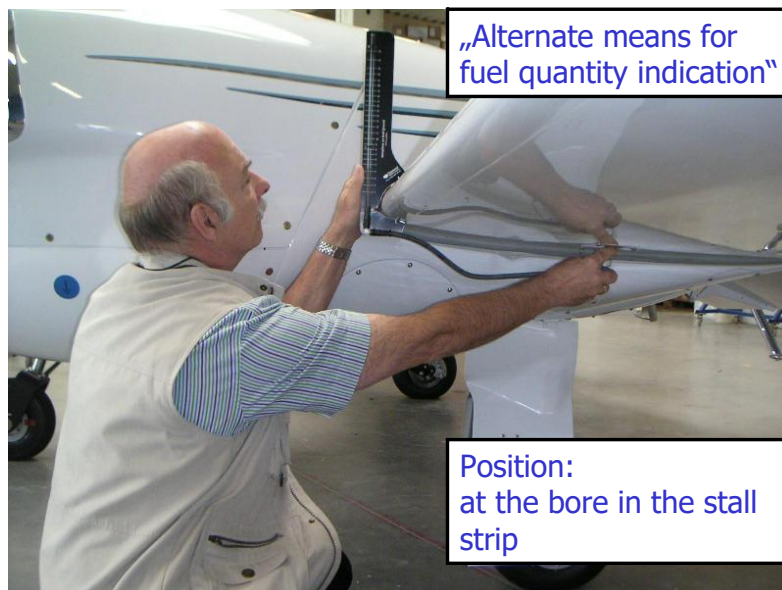
Fuel System



DA40 NG

143

Fuel system



DA40 NG

144

Fuel system

- For exact indication the airplane must stand on horizontal ground with wings level
- Therefore:
 - „Fuel indicator II“ is available for pitch angle measurement
 - Place it on top of fuselage in front of vertical tail
 - Read pitch angle and use table to determine exact fuel quantity

DA40 NG

145

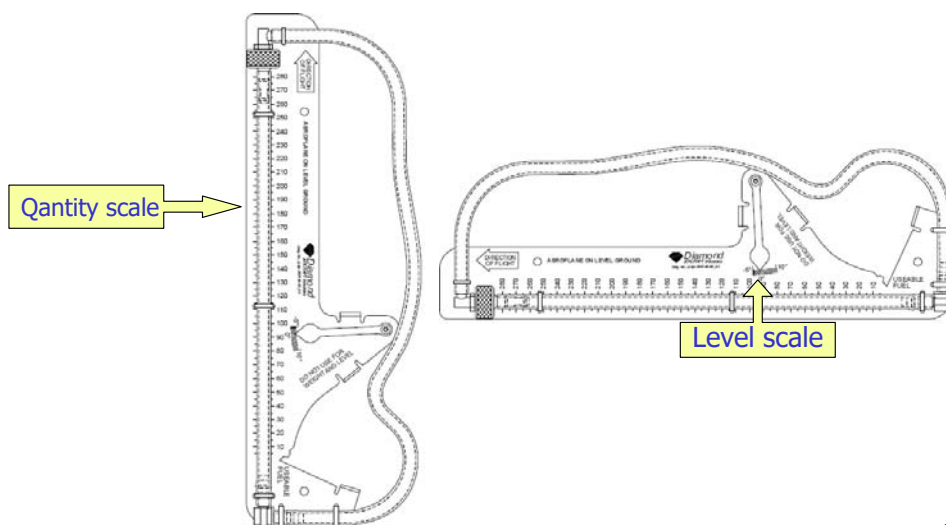
DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel system

Fuel Quantity Indicator II



DA40 NG

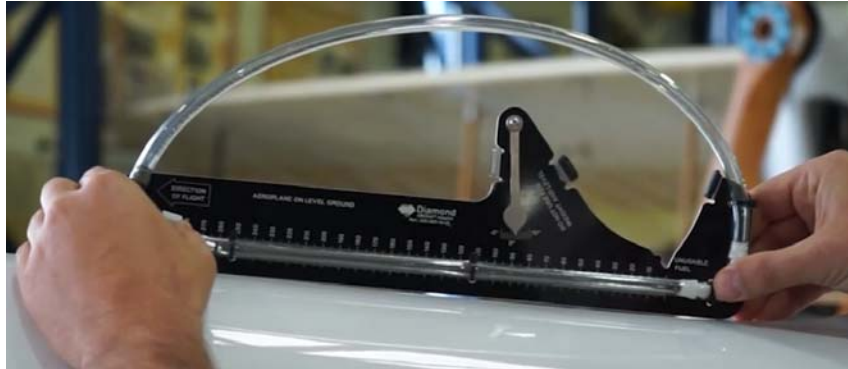
146

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel system



DA40 NG

147

Fuel system

Conversion
tables in the AFM

DA 40 NG AFM **Diamond** Aircraft Temporary Revision
Alternate Means for
Fuel Quantity

The tables are addit:

Standard Tank Configuration

Fuel Quantity Indicator II Pitch Angle Reading					Usable Fuel Quantity	
1°	2°	3°	4°	5°	US gal	Liter
up to 5	up to 5	up to 5	up to 5	up to 5	0	0
25	21	16	12	6	1	3.8
47	37	28	20	10	2	7.6
72	58	45	32	20	3	11.3
83	75	55	40	30	4	15.1
90	87	78	71	55	5	18.9
105	97	90	80	73	6	22.7
112	107	98	82	83	7	26.5
123	115	108	103	95	8	30.3
135	128	120	112	105	9	34.1
148	137	130	124	112	10	37.9
160	152	145	137	130	11	41.6
175	168	167	150	143	12	45.4
192	188	180	172	165	13	49.2
225	215	208	202	192	14	53.0

Doc. # 5.01.15-E TR-MAN-40-815 18-Nov-2015 Page 7-34c

DA 40 NG AFM **Diamond** Aircraft Temporary Revision
Alternate Means for
Fuel Quantity

Long Range Tank (if installed) Configuration

Fuel Quantity Indicator II Pitch Angle Reading					Usable Fuel Quantity	
1°	2°	3°	4°	5°	US gal	Liter
up to 5	up to 5	up to 5	up to 5	up to 5	0	0
35	25	16	8	1	1	3.8
45	35	30	20	15	2	7.6
55	48	40	35	28	3	11.3
75	68	55	47	39	4	15.1
92	80	72	65	55	5	18.9
110	90	78	70	65	6	22.7
118	108	95	87	77	7	26.5
130	123	110	100	90	8	30.3
140	132	115	102	95	9	34.1
148	135	128	122	113	10	37.9
162	148	138	130	118	11	41.6
174	158	150	138	131	12	45.4
180	171	162	156	146	13	49.2
185	180	175	166	156	14	53.0
200	195	184	176	168	15	56.8
217	205	195	188	181	16	60.6
232	220	215	204	195	17	64.4
248	238	230	221	214	18	68.1

Doc. # 5.01.15-E TR-MAN-40-815 18-Nov-2015 Page 7-34d

DA40 NG

148

Fuel vents



DA40 NG

149

Fuel tank drain



DA40 NG

150

Fuel gascolator drain



DA40 NG

151

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel gascolator drain



DA40 NG

152

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel system

- 2 parallel electrical low pressure fuel pumps
- each ECU has an associated fuel pump (ECU A: pump A, ECU B: pump B)
- normal Ops: only one pump working
- when ECU switches over: fuel pumps switch over as well
- When pump fails (low fuel pressure):
 - automatic switch over to other pump by switching to the other ECU
- For TKOF, LDG and with fuel pressure failure:
 - both pumps switched ON manually with FUEL PUMP switch

DA40 NG

153

Fuel pumps



DA40 NG

154

Fuel pumps



CHECK BEFORE ENGINE START	13 Fuel pumps.....OFF	13
BEFORE TAKE OFF CHECK	18 Fuel pumps ON	18
CLIMB TO CRUISE CHECK	1 Flaps CHECKED UP	1
	2 Fuel pumps OFF	2
DESCENT / APPROACH CHECK	8 Fuel pumps ON	8
AFTER LANDING CHECK	3 Fuel pumps.....OFF	3

DA40 NG

155

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel pressure warning



- Warning annunciation:
FUEL PRESS

DA40 NG

156

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel pressure warning



FUEL PRESS

FUEL PRESSURE LOW

- Check fuel quantity
- Check fuel valve NORMAL
- Switch fuel pumps ON
 - If FUEL PRESS warning remains:
 - ⇒ Fuel valve to EMERGENCY
 - ⇒ Switch fuel pumps OFF
 - If FUEL PRESS warning still remains
 - ⇒ Be prepared for engine fail

DA40 NG

157

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner



Intentionally blank

DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG



159

DA40NG Diesel Operation

The following item is added to the Modifications checklist:

Modification	Source	Installed	
Diesel Operation	OÄM 40-370	☐ yes	☐ no

9.2 LIST OF SUPPLEMENTS

The following item is added:

Airplane S/N:		Registration:		Date:	
Sup. No.	Title	Rev. No.	Date	applicable	
				YES	NO
O04	Diesel Operation	0	06-Dec-2013	☐	☐

DA40 NG

160

DA40NG Diesel Operation



■ Fuel Temperature

■ Diesel Operation

- Minimum: -30° C
- Maximum: 60° C

■ Modified G1000 Software:

Indi- cation	Red arc/bar = lower prohibited range	Yellow arc/bar = caution range	Green arc/bar = normal operating range	Yellow arc/bar = caution range	Red arc/bar = upper prohibited range
Fuel temp.	below -30°C	-30° to +4°C	+5° to 55°C	55° to 60°C	above 60°C

DA40 NG

161

DA40NG Diesel Operation



■ Operational Limitations (Fuel Temperature):

No engine start below	No take-off below	Diesel Fuel Class
-5°C (+23°F)	+5°C (+41°F)	Diesel Fuel of unknown class or unknown fuel blend
-10°C (+14°F)	-5°C (+23°F)	Diesel Fuel Class C
-15°C (+5°F)	-10°C (+14°F)	Diesel Fuel Class D, E or F

DA40 NG

162

DA40NG Diesel Operation



Diesel Fuel Classes (EN 590)

(„Temperate“ climatic zones)

Property	Unit	Limits					
		Class A	Class B	Class C	Class D	Class E	Class F
CFPP	° C	+5	0	-5	-10	-15	-20
CFPP	° F	+41	+32	+23	+14	+5	-4

CFPP Cold filter plugging point

„Winter Diesel“: Class F

„Arctic“ climatic zones: other classes

DA40 NG

163

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

DA40NG Diesel Operation



- If JET Fuel is used, make sure that no Diesel Fuel is remaining in the tanks, neither in the left nor in the right tank.
- Otherwise the temperature limitations for Diesel Fuel operation must be observed.
- **The fuel filter is not heated!**

DA40 NG

164

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

DA40NG Diesel Operation



- If the airplane is operated in a cold environment, it must be changed from Diesel Fuel operation to JET Fuel operation.
- To ensure that no blend of JET Fuel with Diesel Fuel is in one of the tanks, each tank must be refilled at least twice with more than 17.2 US gal (65 l) of JET Fuel. Otherwise both tanks must be drained before refueling with JET Fuel.

NOTE

- In order to provide information about the fuel grade it is recommended to enter the fuel grade in the airplane log each time fuel is refilled.

DA40 NG

165

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

DA40NG Diesel Operation



Performance below 10° C OAT

TKOF Ground Roll	Add 6%
TKOF Distance	
TKOF Climb	Reduce by 50 ft/min
Cruise Climb	

DA40 NG

166

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG

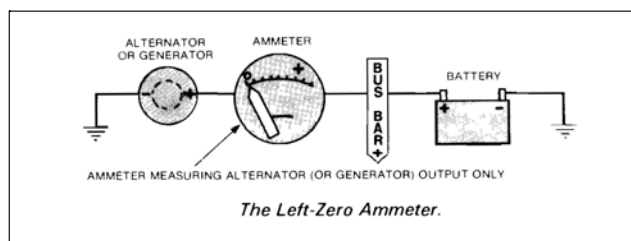
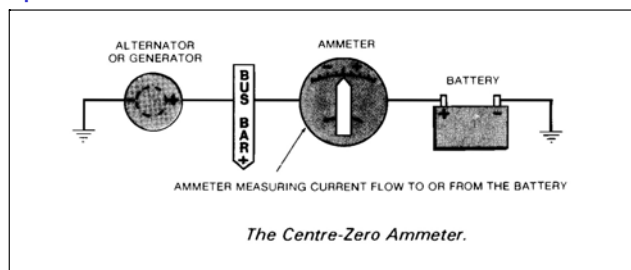


167

Electrical system

General information:

2 types of amperemeter can be installed in aircraft:

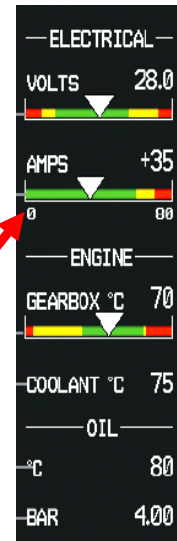
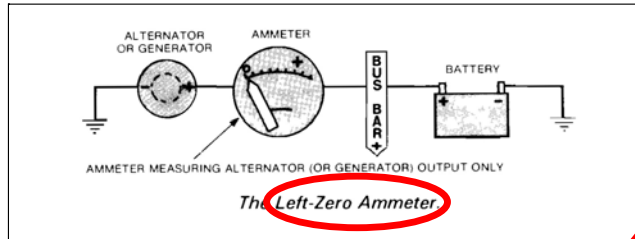


DA40 NG

168

Observe!

The Amperemeter on the G1000 is a



DA40 NG

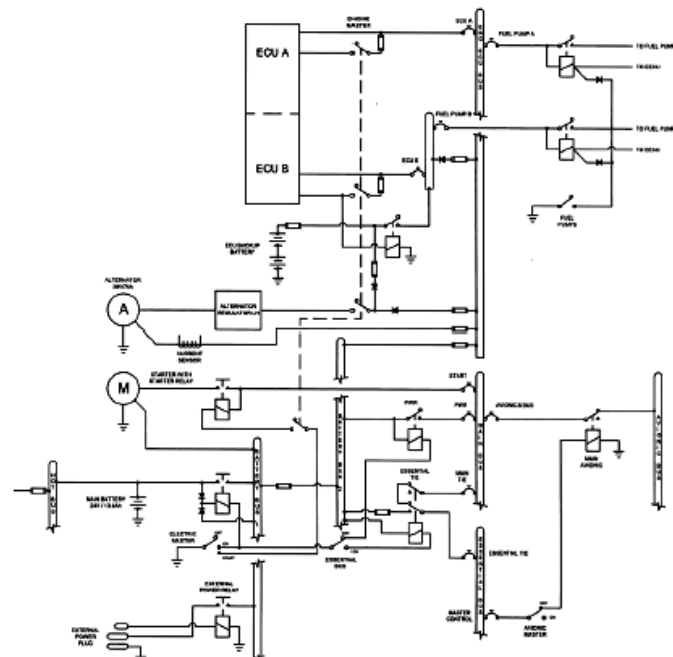
169

© Diamond Flight Training

169

Compiled by Peter Schmidleitner

Diamond
FLIGHT TRAINING



DA40 NG

170

© Diamond Flight Training

Compiled by Peter Schmidleitner

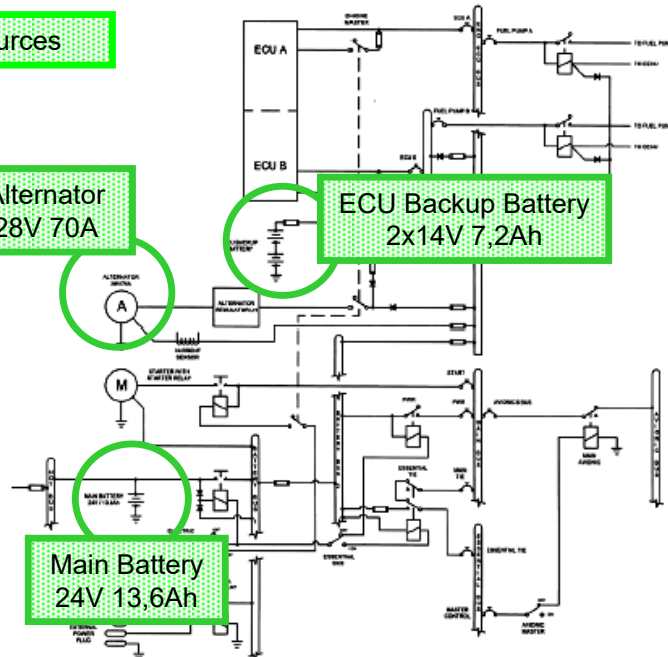
Electrical system

Power sources

Alternator
28V 70A

ECU Backup Battery
2x14V 7.2Ah

Main Battery
24V 13,6Ah



DA40 NG

171

© Diamond Flight Training

Compiled by Peter Schmidleitner

Electrical system

Power distribution

ENG ECU Bus

Battery Bus 2

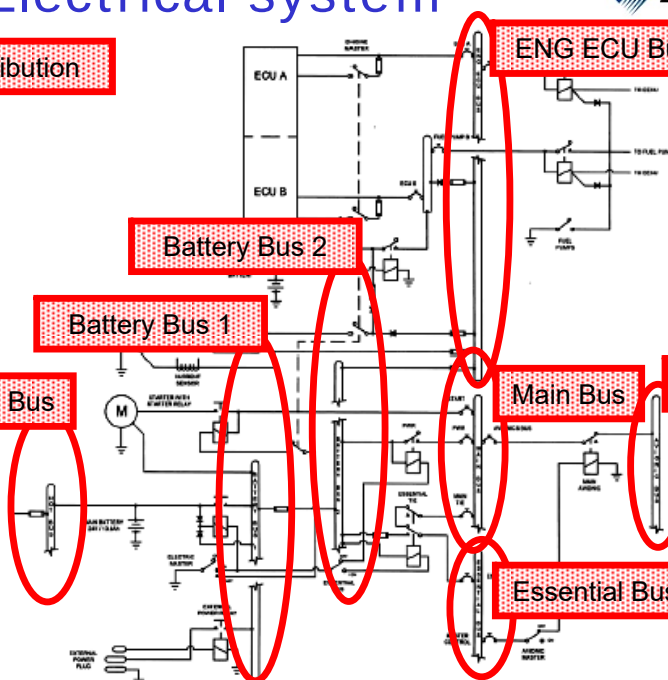
Battery Bus 1

Hot Battery Bus

Main Bus

Avionic Bus

Essential Bus

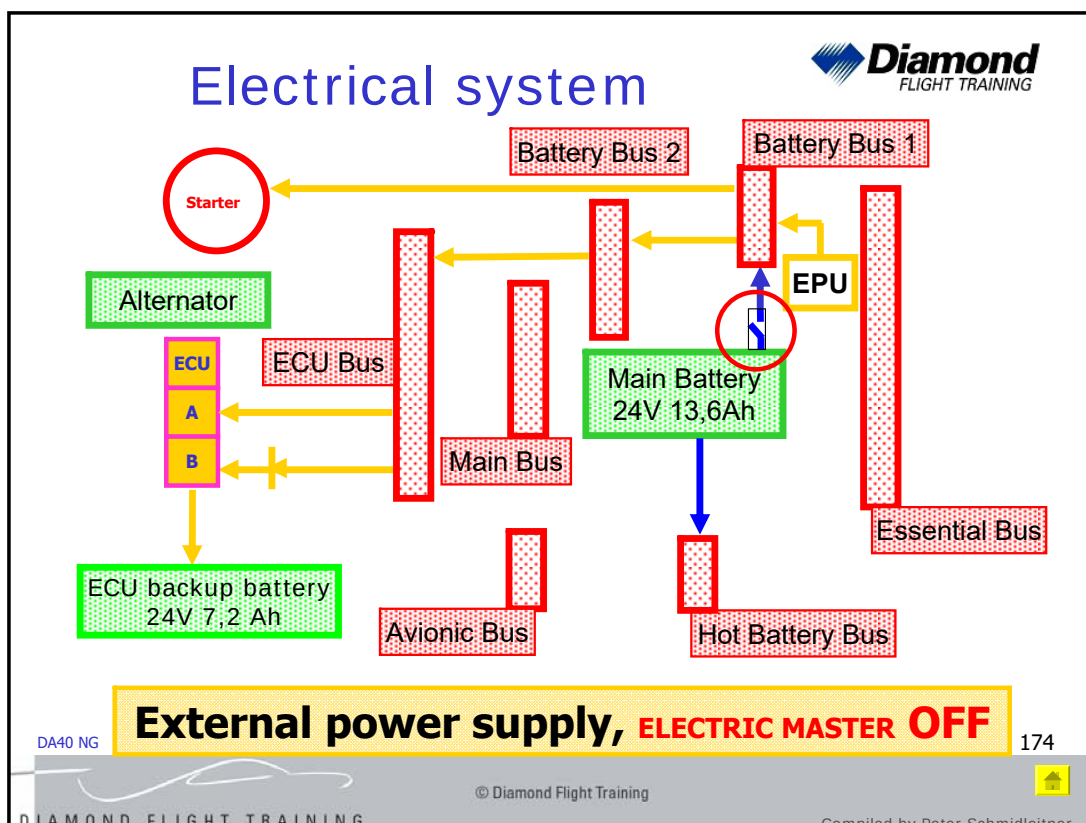
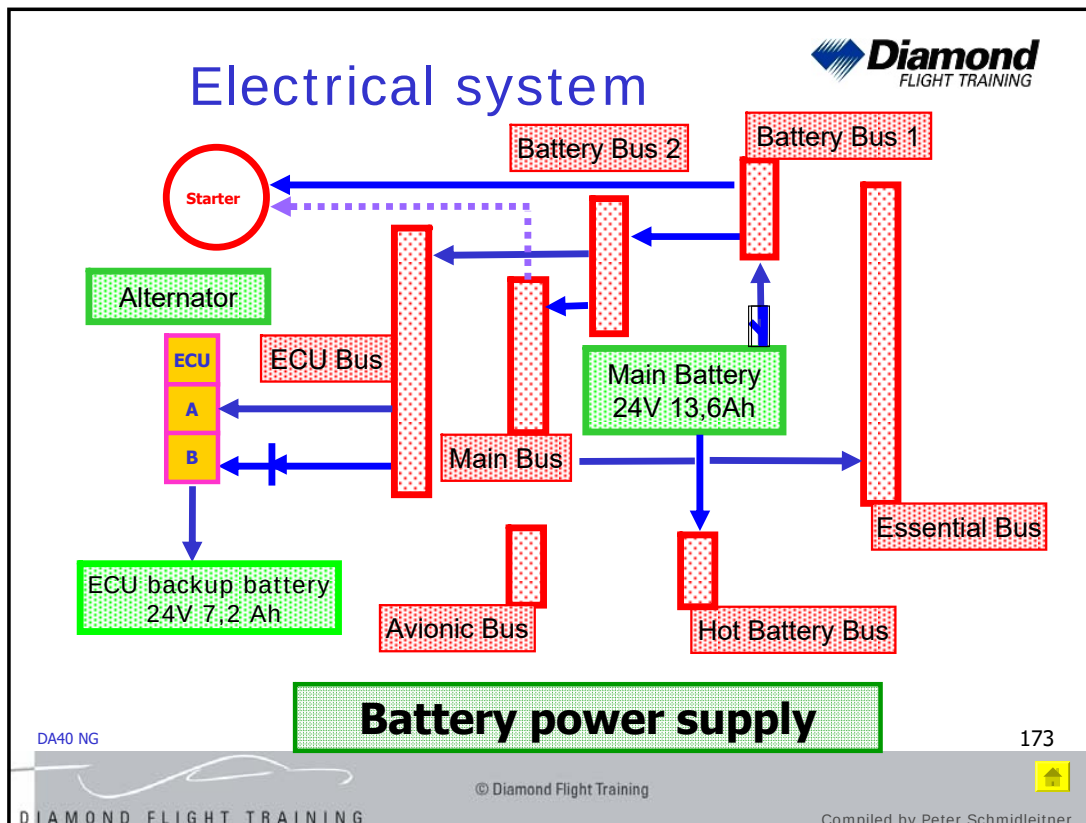


DA40 NG

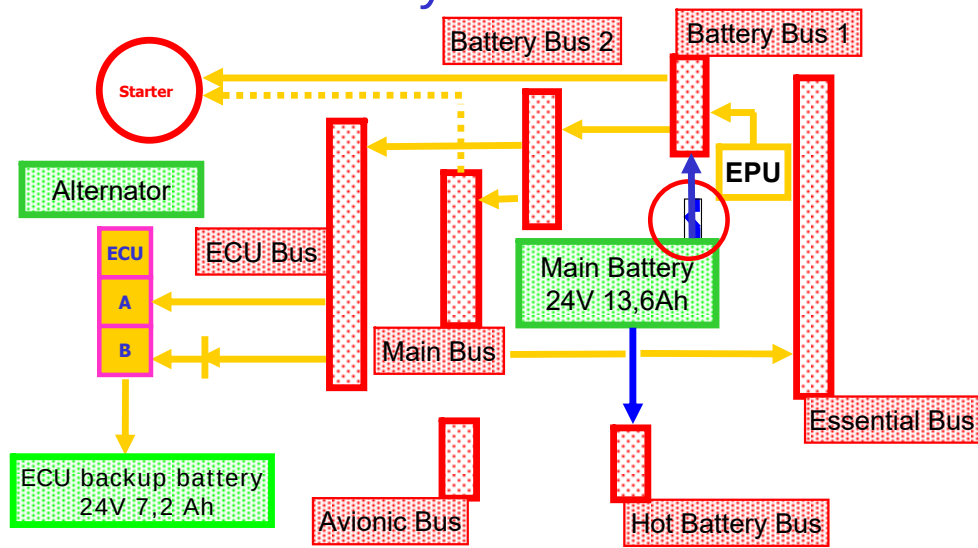
172

© Diamond Flight Training

Compiled by Peter Schmidleitner



Electrical system



External power supply, ELECTRIC MASTER ON

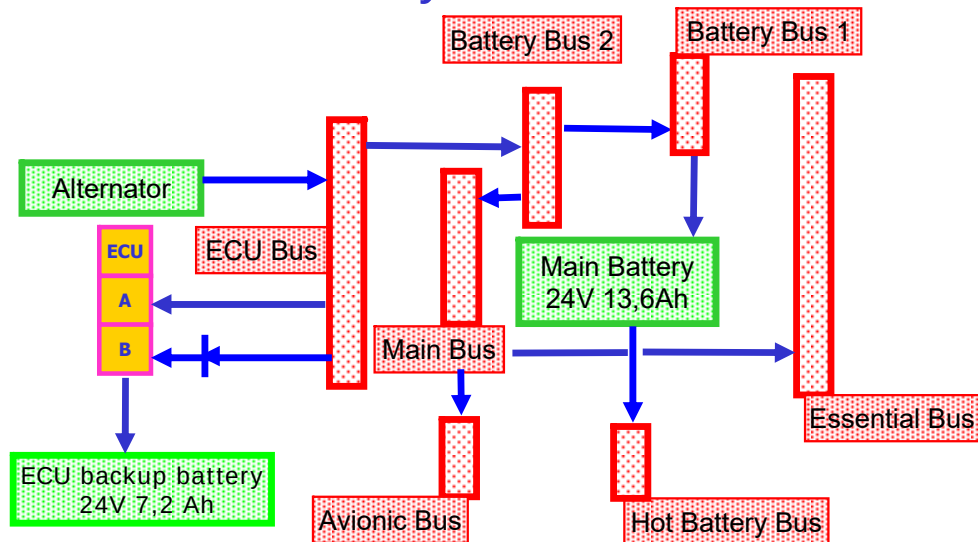
DA40 NG

175

© Diamond Flight Training

Compiled by Peter Schmidleitner

Electrical system



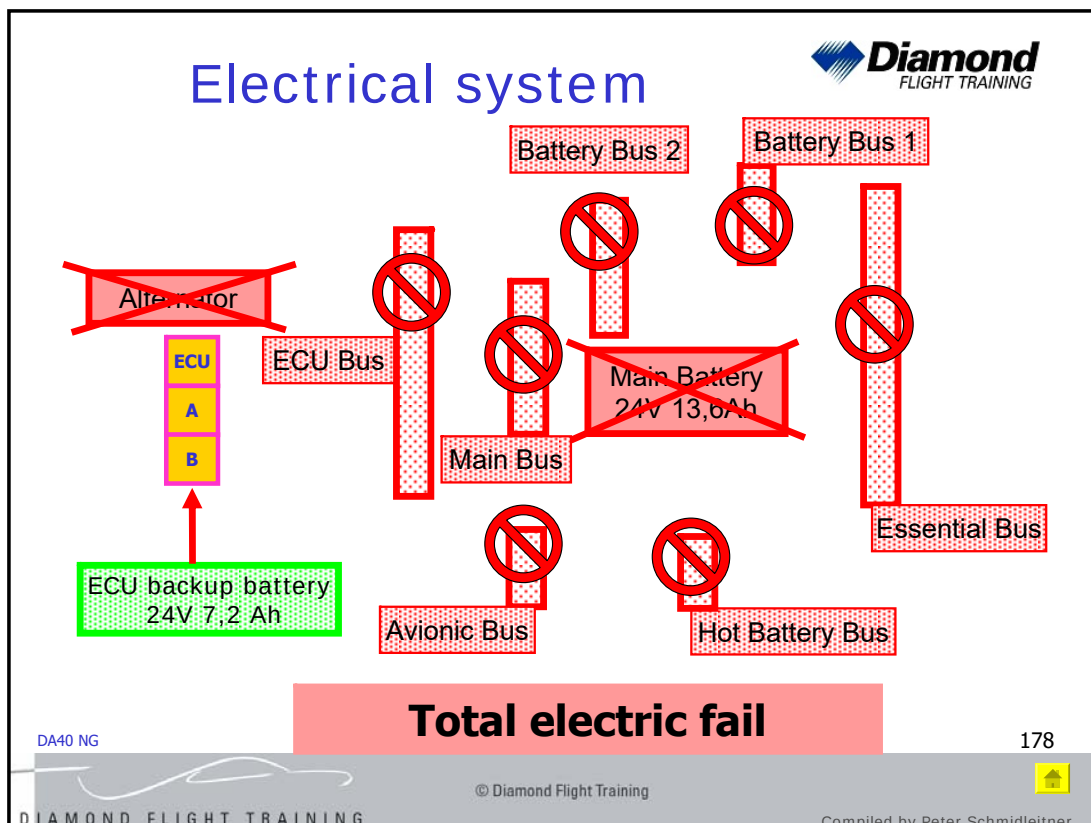
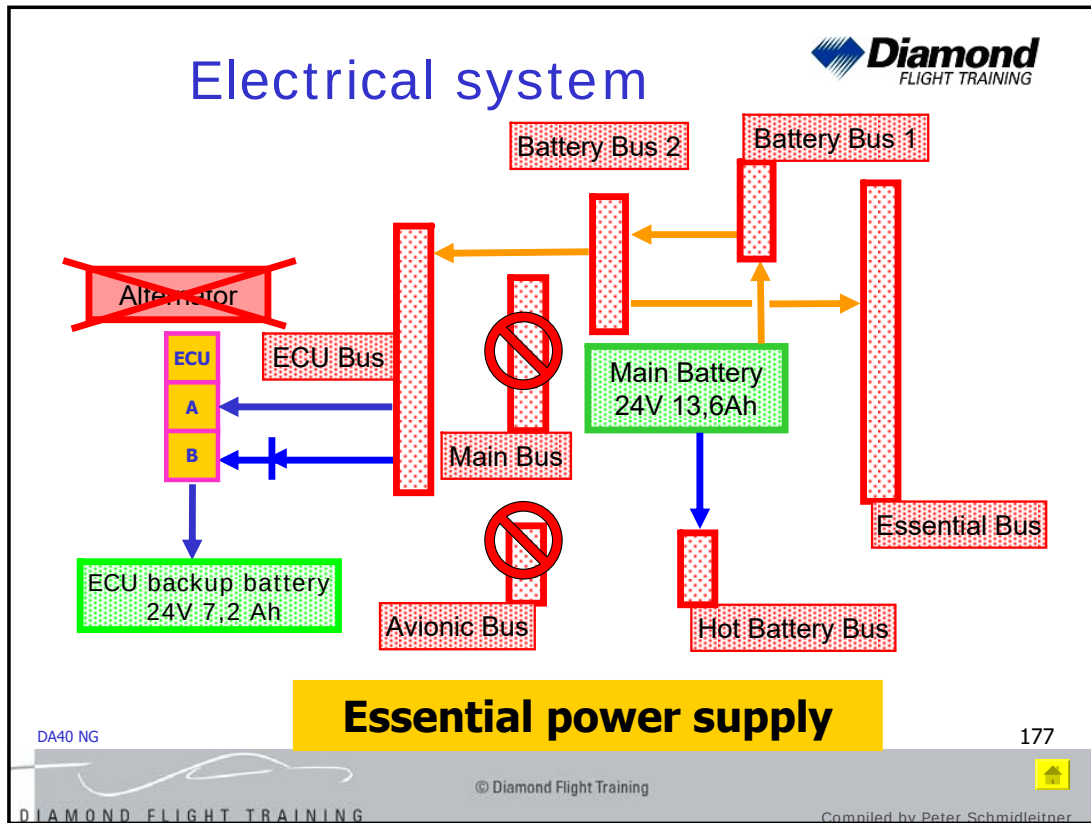
Normal power supply

DA40 NG

176

© Diamond Flight Training

Compiled by Peter Schmidleitner



Hot battery bus



- Accessory power plug
- ELT

DA40 NG

179

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Battery bus 1 and 2



- Battery Bus 1
 - starter heavy duty power
- Battery Bus 2
 - ECU BUS
 - Main Bus
(Electric Master ON, ESS BUS OFF)
 - Essential Bus
(Electric Master ON, ESS BUS ON)

DA40 NG

180

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Main bus

- Essential Bus
(ESS BUS OFF)
- Avionic Bus
- Starter control
- MFD
- Fuel x-fer pump
- Avionic-, CDU-cooling fans
- Instrument lights
- Taxi + map lights
- Position lights
- Strobe lights

DA40 NG

181

Essential bus

- PFD
- Horizon
- Air Data Computer
- AHRS
- COM 1
- GPS/NAV 1
- Flood light
- Landing light
- Transponder
- Pitot heat
- Flap system
- Engine instruments

DA40 NG

182

Avionic bus

- COM 2
- GPS/NAV 2
- Audio panel
- Autopilot
- DME
- ADF
- WX 500
- TAS

DA40 NG

183

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Essential Bus, Avionic Master Electric Master



DA40 NG

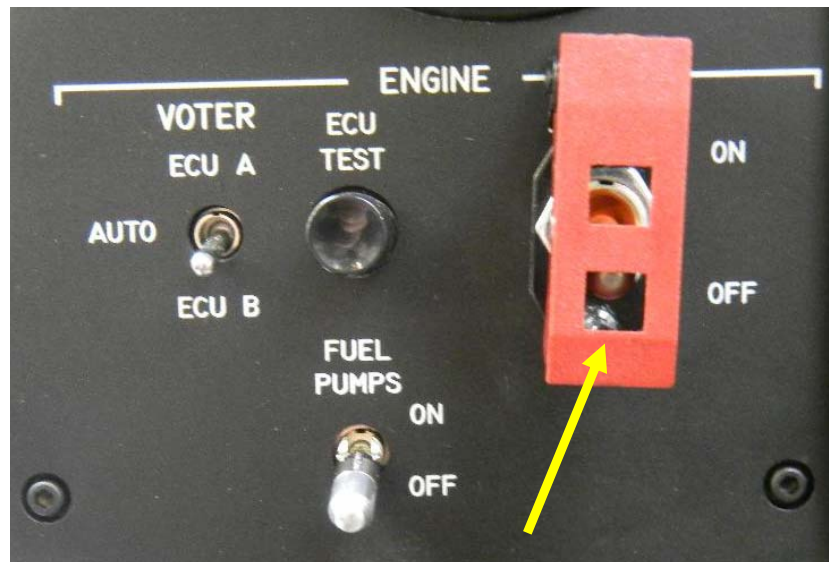
184

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Engine Master



DA40 NG

185

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

External Power Receptacle



DA40 NG

186

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

External Power Receptable



DA40 NG

187

External Power Receptable



DA40 NG

188

Essential bus switch



The mysterious Essential Bus Switch

DA40 NG

189

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Switch location



DA40 NG

190

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

When is it used ?

- When the alternator fails
- to disconnect unnecessary electrical consumers
- to supply battery power to essential electrical consumers

DA40 NG

191

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Essential Electrical Power

- **Main unserviceable systems**
 - Starter control
 - MFD
 - Fuel x-fer pump
 - Avionic-, CDU-cooling fans
 - Instrument lights
 - Taxi + map lights
 - Position lights
 - Strobe lights
 - Avionics Bus
 - COM2, NAV/GPS2, Audio, Autopilot, DME, ADF, WX500, TAS

DA40 NG

192

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Autopilot GFC700



DA40 NG

193

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS

- „AFCS“:
 - Automatic Flight Control System
 - Flight Director
 - Autopilot

DA40 NG

194

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS



DA40 NG

195

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



DA40 NG

196

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS

- GFC 700 functions:
 - Flight Director (FD)
 - AUTO Pilot (AP)
 - Manual Electric Trim (MET)

DA40 NG

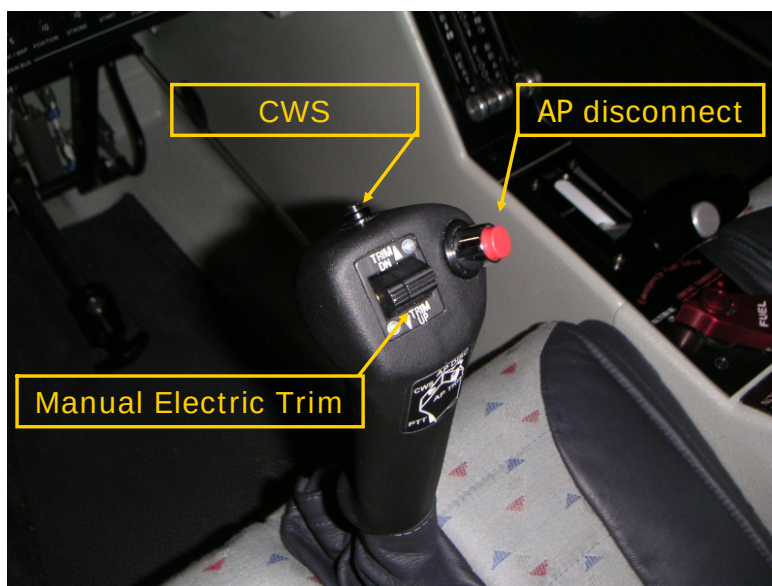
197

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Autopilot



DA40 NG

198

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

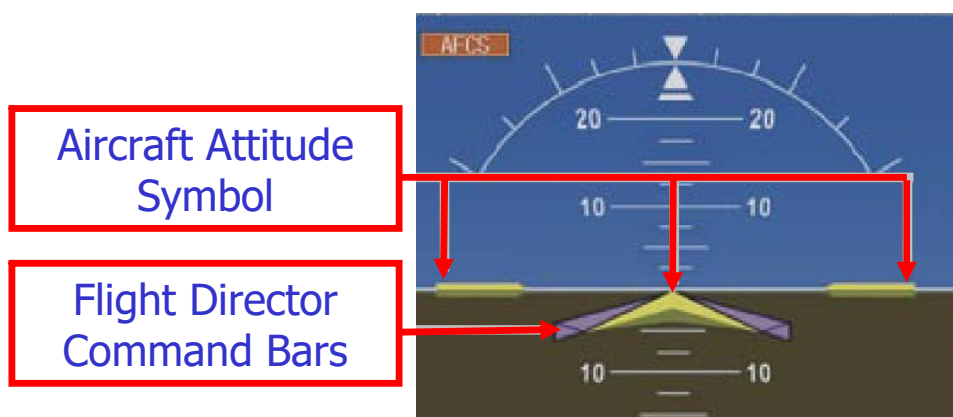
Autopilot



DA40 NG

199

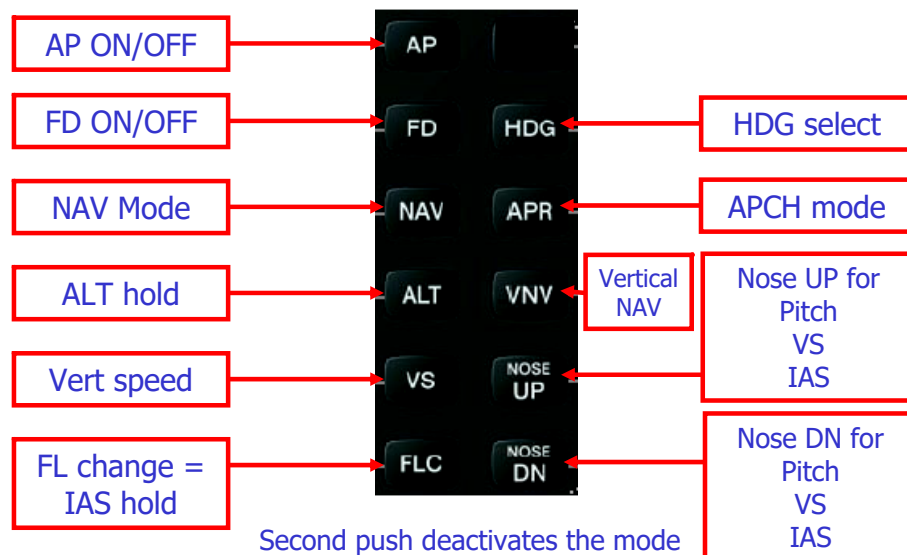
GFC 700 AFCS



DA40 NG

200

GFC 700 AFCS



DA40 NG

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved.
© Diamond Flight Training

201

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS

- FLC („Flight Level Change“ Mode):
 - „FLC“ = IAS hold
 - A target altitude must be selected first
 - FLC does **not initiate** the level change to the selected altitude
 - Current altitude is maintained until either engine power or IAS reference are adjusted
 - FLC does not allow the aircraft to climb or descend away from the target altitude

DA40 NG

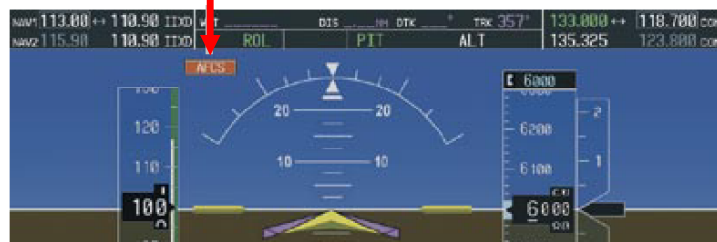
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS

System Status Annunciation



DA40 NG

203

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS

Condition	Annunciation	Description
Pitch Failure	PTCH	Pitch axis control failure. AP is inoperative.
Roll Failure	ROLL	Roll axis control failure. AP is inoperative.
AP TRIM Switch Stuck, or Pitch Trim Axis Control Failure	PTRM	If annunciated when AP is engaged, a failure has occurred in the pitch trim system. If annunciated when AP is not engaged, a failed or stuck switch is causing the annunciation.
System Failure	AFCS	AP and manual electric trim (AP TRIM) are unavailable. FD may still be available.
Elevator Mistrim Up	↑ELE	A condition has developed causing the pitch servo to provide a sustained force in the nose up direction.
Elevator Mistrim Down	↓ELE	A condition has developed causing the pitch servo to provide a sustained force in the nose down direction.
Aileron Mistrim Left	←AIL	A condition has developed causing the roll servo to provide a sustained left force.
Aileron Mistrim Right	AIL→	A condition has developed causing the roll servo to provide a sustained right force.
Preflight Test	PFT	Performing preflight system test. Upon completion of the test, the aural alert will be heard.
	PFT	Preflight system test has failed.

DA40 NG

204

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



When FD or AP is engaged:

Bank Angle	Flight Director Response
$< 6^\circ$	Rolls wings level
6° to 25°	Maintains current aircraft roll attitude
$> 25^\circ$	Limits bank to 25°

DA40 NG

205

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



DA40 NG

206

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



DA40 NG

207

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

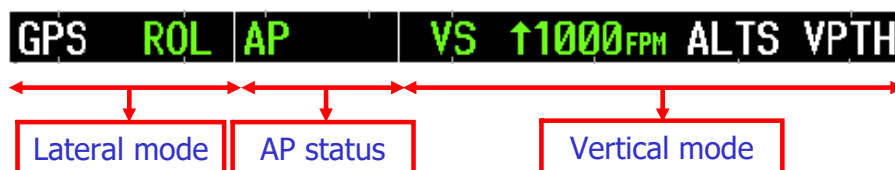
DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Armed modes:

white

Active modes:

green

DA40 NG

208

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

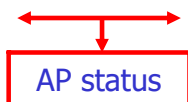
Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“

GPS ROL AP VS ↑1000FPM ALTS VPTH



Automatic disconnect



Manual disconnect



Control Wheel Steering

DA40 NG

209

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

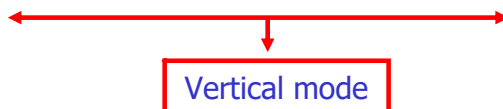
Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“

GPS ROL AP VS ↑1000FPM ALTS VPTH



DA40 NG

210

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Vertical modes

Vertical Mode	Control	Annunciation
Pitch Hold	(default)	PIT
Selected Altitude Capture	*	ALTS
Altitude Hold	ALT Key	ALT nnnnn FT
Vertical Speed	VS Key	VS nnnn FPM
Flight Level Change	FLC Key	FLC nnn KT
Vertical Path Tracking	VNV Key	VPTH
VNV Target Altitude Capture	**	ALTV
Glidepath***	APR Key	GP
Glideslope		GS
Go Around	GA Button	GA

DA40 NG

211

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS



Vertical mode range and change increments

Pitch Mode	Control	Annunciation	Reference Range	Reference Change Increment
Pitch Hold	(default)	PIT	-15° to +20°	0.5°
Altitude Hold	ALT Key	ALT nnnnn FT		
Vertical Speed	VS Key	VS nnnn FPM	-3000 to +1500 fpm	100 fpm
Flight Level Change	FLC Key	FLC nnn KT	70 to 165 kt	1 kt
Glideslope Arm/Capture/Track	APR Key	GS		
Go Around	GA Button	GA		

DA40 NG

212

DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



DA40 NG

213

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Roll Mode	Control	Annunciation
Roll Hold	(default)	ROL
Heading Select	HDG Key	HDG
Navigation, GPS Arm/Capture/Track	NAV Key	GPS
Navigation, VOR Enroute Arm/Capture/Track		VOR
Navigation, LOC Arm/Capture/Track (No Glideslope)		LOC
Navigation, Backcourse Arm/Capture/Track		BC
Approach, GPS Arm/Capture/Track	APR Key	GPS
Approach, VOR Arm/Capture/Track		VAPP
Approach, ILS Arm/Capture/Track (Glideslope Mode automatically armed)		LOC
Go Around	GA Button	GA

DA40 NG

214

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS

GA mode active

GA AP GA ALTS

- Wings level
- 7° pitch up

DA40 NG

215

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS



DA40 N DAT 71°F

-16

DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

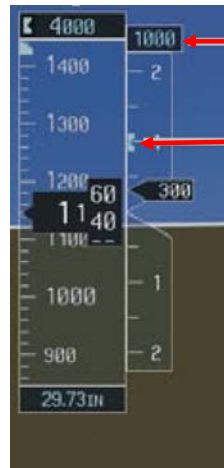
© Diamond Flight Training

Compiled by Peter Schmidleitner

GFC 700 AFCS



Vertical Speed Mode



Selected Vertical Speed

Vertical Speed Bug

DA40 NG

217

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



IAS Hold Mode („Flight Level Change“)

Selected Airspeed

Airspeed Reference Bug



DA40 NG

218

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



Overspeed Protection



MAXSPD

flashing

DA40 NG

219

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved.
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

GFC 700 AFCS



LIMITATIONS FOR GFC 700 AUTOPILOT SYSTEM:

DO NOT USE AP IF "ALTERNATE STATIC" IS OPEN.

CONDUCT AP AND TRIM CHECK PRIOR TO EACH FLIGHT (SEE AFM).

AUTOPILOT OFF DURING TAKE-OFF AND LANDING.

MAXIMUM SPEED FOR AUTOPILOT OPERATION IS 165 KIAS.
MINIMUM SPEED FOR AUTOPILOT OPERATION IS 70 KIAS.

MINIMUM ALTITUDE FOR AUTOPILOT OPERATION:

CRUISE, CLIMB, DESCENT AND MANEUVERING: 800 FEET AGL
APPROACH: 200 FEET AGL

DA40 NG

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

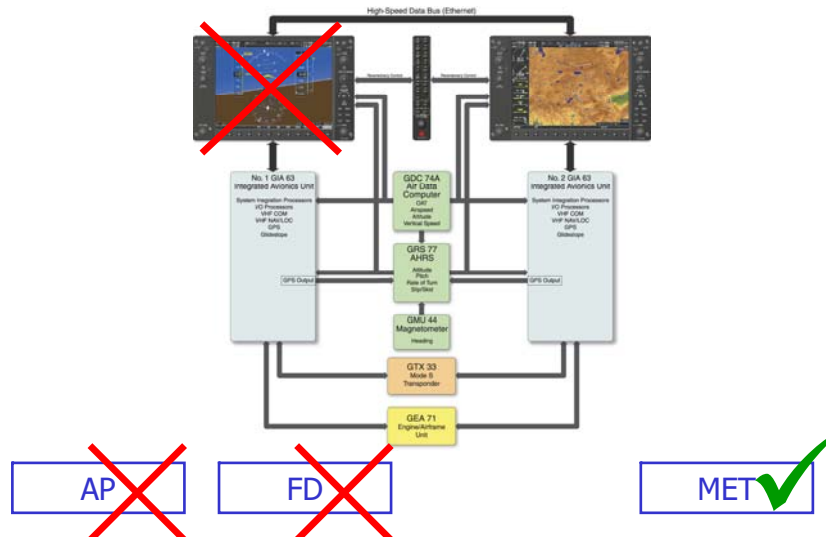
G 1000 System Loss Effect upon Autopilot Operation

AHRS	The Autopilot disconnects, Autopilot and Flight Director are inoperative. Manual electric trim is available.
HDG function of AHRS	The Autopilot will remain engaged with the loss of the HDG Mode.
MFD	The Autopilot will remain engaged with limited functionality.
PFD	The Autopilot disconnects, Autopilot and Flight Director are inoperative. Manual electric trim is available.
GIA No. 1	The Autopilot disconnects and Autopilot, Flight Director and manual electric trim are inoperative.
GIA No. 2	The Autopilot disconnects and Autopilot and manual electric trim are inoperative. Flight Director is available.
GPS No. 1 and 2	The Autopilot and Flight Director operates in NAV modes only (LOC, BC, VOR, VAPP) with reduced accuracy.
ADC	The Autopilot disconnects and Autopilot is inoperative. The flight director is available except for air data modes (ALT, VS, FLC). Manual electric trim is available.

DA40 NG

221

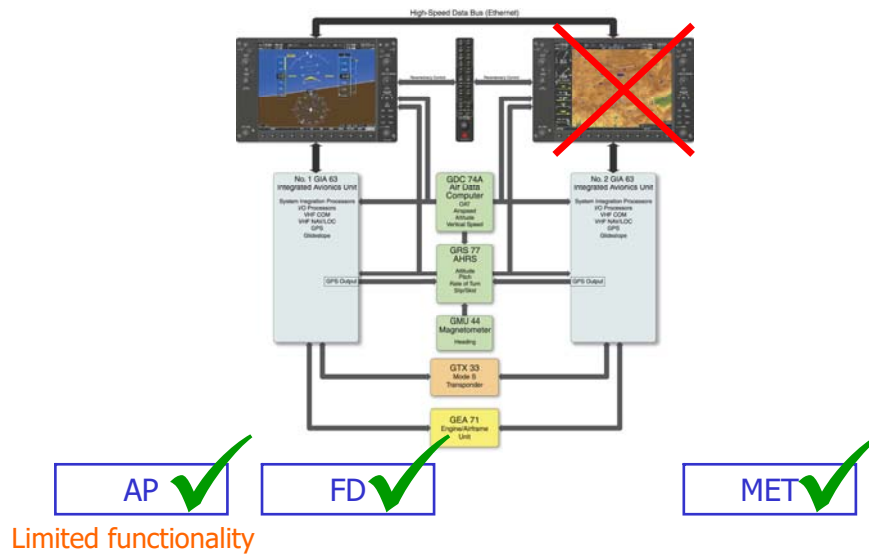
G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

222

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

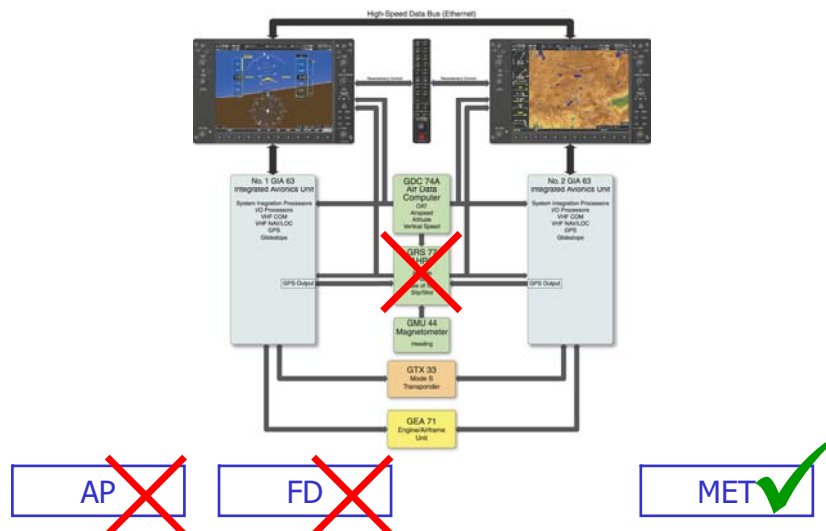
223

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

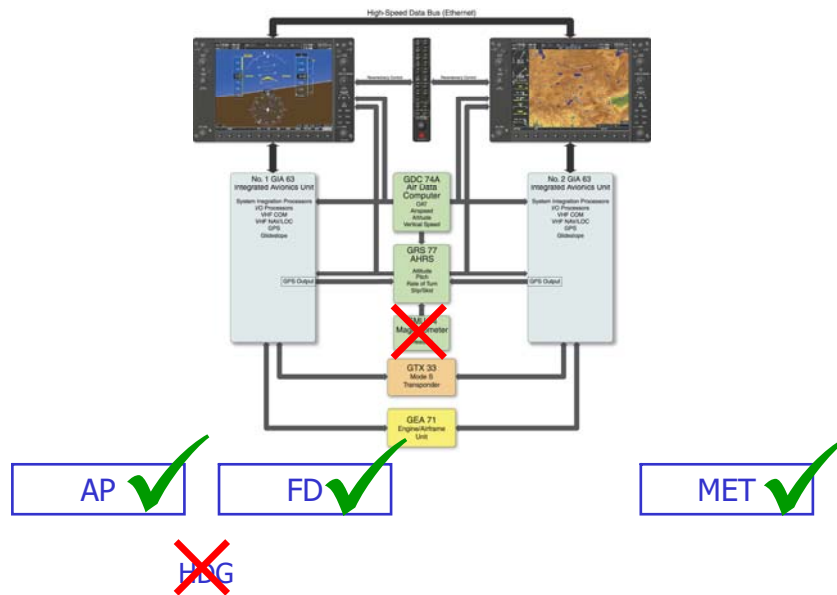
224

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

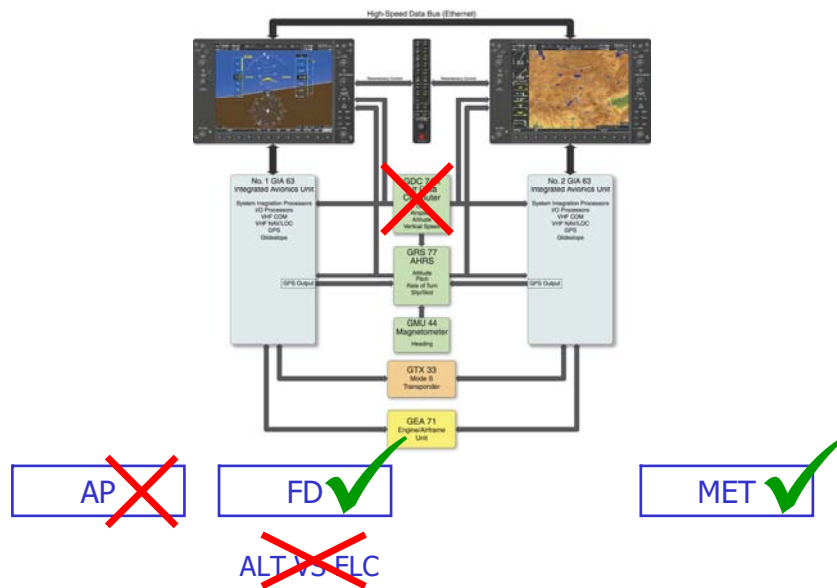
225

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

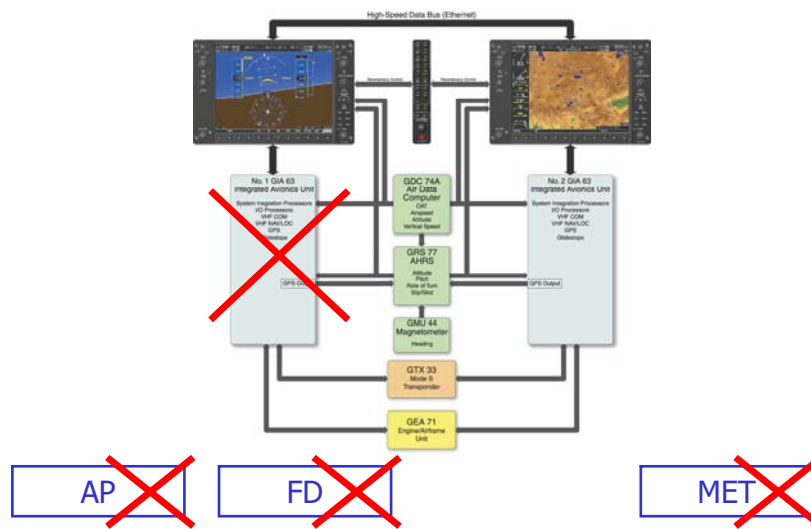
226

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

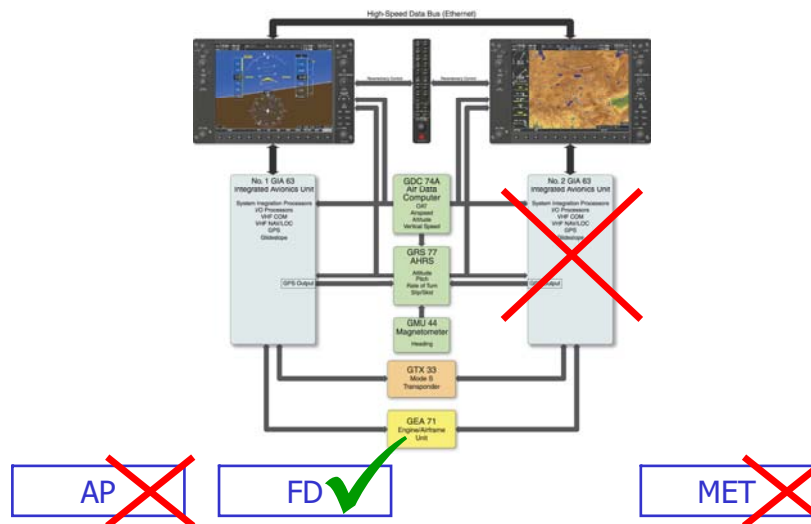
227

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

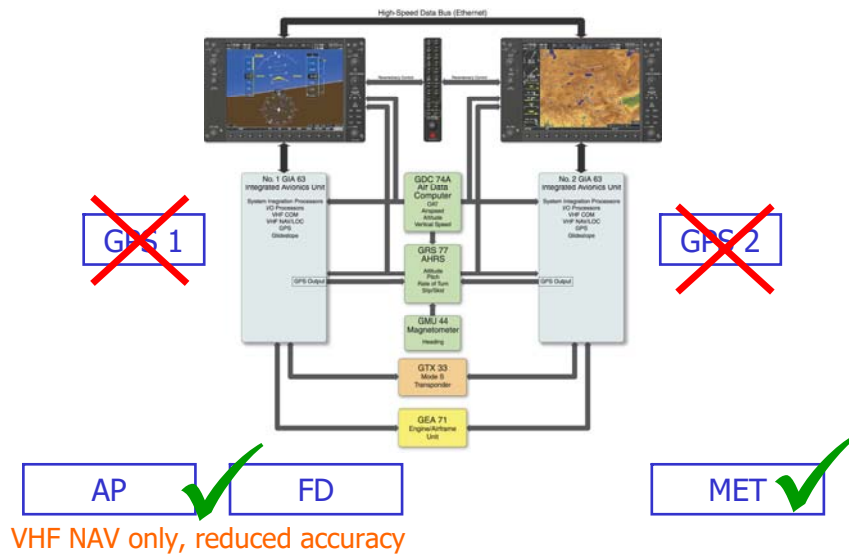
228

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

G 1000 System Loss Effect upon Autopilot Operation



DA40 NG

229

© Diamond Flight Training

Compiled by Peter Schmidleitner

Intentionally blank

DA40 NG

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG



231

TKOF, LDG Performance general

For temperatures, altitudes and weights between those provided, use a linear interpolation between the neighboring values.

For weights below 1100 kg (2425 lb), use data for the lowest weight.

For operation at outside air temperatures lower than provided in these tables, use data for lowest temperature shown.

Use extreme caution for operation at outside air temperatures higher than provided in the tables (areas are indicated with a diagonal line).

Effect of wind	The effect of 50% of the headwind component and 150% of the tailwind component is already incorporated in the head and tailwind factors.		
	TKOF/LDG distance	TKOF	LDG
Headwind	minus 10 % for each	12 kt	20 kt
Tailwind	plus 10 % for each	2 kt	3 kt

Effect of slope		TKOF	LDG
Uphill	For each 1% increase ground roll by	15%	
Downhill			10%

DA40 NG

232

X-wind

Max demonstrated crosswind component:
25 kts

DA40 NG

233

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Performance Tables

- Fuel Flow
- Stalling Speeds
- TKOF Ground Roll / TKOF Distance
- Climb Performance
 - Take off climb
 - Cruise climb
- Time, Fuel, Distance to Climb
- Cruise Performance
- LDG Ground Roll / LDG Distance
 - Flaps LDG
 - Abnormal Flap Position
- Go-Around Climb Performance

DA40 NG

234

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Fuel Flow

Fuel Flow		
Power Setting [%]	Fuel Flow [US gal / h]	Fuel Flow [Liter / h]
30	2.9	11.0
35	3.3	12.5
40	3.7	14.0
45	4.0	15.5
50	4.4	16.5
55	4.7	18.0
60	5.1	19.5
65	5.6	21.5
70	6.1	23.0
75	6.6	25.0
80	7.1	27.0
85	7.6	28.5
90	8.1	30.5
92	8.3	31.5
100	9.4	35.5

6,6 USG/hr

DA40 NG

235

Stalling Speeds

1000 kg (2205 lb)	Bank Angle							
	0°		30°		45°		60°	
Flaps	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	58	56	59	60	64	66	76	79
T/O	54	53	58	57	63	63	75	74
LDG	55	52	56	55	61	61	72	73

1100 kg (2425 lb)	Bank Angle							
	0°		30°		45°		60°	
Flaps	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	61	59	63	64	70	71	83	84
T/O	56	55	60	60	66	66	79	78
LDG	57	54	59	58	65	65	77	77

1200 kg (2646 lb)	Bank Angle							
	0°		30°		45°		60°	
Flaps	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	64	61	67	66	73	73	86	87
T/O	60	57	64	62	69	68	82	81
LDG	59	56	62	61	68	67	81	80

1310 kg (2888 lb)	Bank Angle							
	0°		30°		45°		60°	
Flaps	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS	KIAS	KCAS
UP	66	63	68	68	74	75	88	89
T/O	62	59	65	63	71	70	84	83
LDG	60	58	63	62	69	69	82	82

DA40 NG

236

Take off distance



Take-Off Distance - Normal Procedure - 1310 kg / 2888 lb								
Weight: 1310 kg / 2888 lb			Flaps: T/O					
V _R : 67 KIAS			Power: MAX					
V ₅₀ : 72 KIAS			Runway: dry, paved, level					
Press. Alt. [ft] / [m]	Distance [m]	Outside Air Temperature - [°C] / [°F]						ISA
		0 / 32	10 / 50	20 / 68	30 / 86	40 / 104	50 / 122	
SL	Ground Roll	365	385	410	430	460	495	397
	15 m / 50 ft	550	580	610	640	680	720	590
1000 305	Ground Roll	390	410	435	465	500	535	418
	15 m / 50 ft	580	610	640	680	730	770	616
2000 610	Ground Roll	415	440	465	500	540	575	439
	15 m / 50 ft	610	640	680	730	780	830	646
3000 914	Ground R							
	15 m / 50							
4000	Ground R							
	15 m / 50							

Available tables:

1310 kg
1280 kg
1200 kg
1100 kg

Up to 10000 ft

Headwind: TKOF distance minus 10% for each 12 kt
Tailwind: TKOF distance plus 10% for each 2 kt
Uphill slope: Ground roll plus 15% for each 1% of slope
Without wheel fairings: Ground roll plus 20 m
TKOF distance plus 30 m

DA40 NG

237

TKOF on Grass RWY



Length of grass	TKOF roll	Wet grass
5 cm	+ 10%	additional + 20%
5 - 10 cm	+ 30%	
10 - 25 cm	+ 45%	
> 25 cm	TKOF should not be attempted	
+ 50% on soft ground !		

DA40 NG

238

Climb performance

Take - Off Climb - Flaps T/O										
Flaps: T/O		Power: 92% or max. 2100 RPM								
v _y : 72 KIAS										
Weight [kg] / [lb]	Press. Alt. [ft]	Press. Alt. [m]	Rate of Climb - [ft/min]							
			Outside Air Temperature - [°C] / [°F]							
			-20 -4	-10 14	0 32	10 50	20 68	30 86	40 104	50 122
1310 / 2888	SL		660	650	640	630	620	615	590	550
	2000	610	640	630	620	610	605	595	555	515
	4000	1219	620	610	600	595	585	560	520	475
	6000	1829	600	590	580	570	555	520	475	435
	8000	2438	580	570	555	540	525	480	435	395
	10000	3048	555	540	525	510	480	435	395	355
	12000	3658	525	510	495	480	435	400	360	320
	14000	4267	500	485	475	460	425	360	320	280
	16000	4877	490	470	440	385	325	280	240	200
	16400	4999	475	450	420	370	305	260	220	180
SL			675	665	655	645	635	625	600	560

Available tables:

Take-Off Climb,
Flaps T/O

1310 kg
1280 kg
1200 kg
1100 kg

DA40 NG

239

Climb performance

Cruise Climb - Flaps UP										
Flaps: UP		Power: 92% or max. 2100 RPM								
v _y : 88 KIAS										
Weight [kg] / [lb]	Press. Alt. [ft]	Press. Alt. [m]	Rate of Climb - [ft/min]							
			Outside Air Temperature - [°C] / [°F]							
			-20 -4	-10 14	0 32	10 50	20 68	30 86	40 104	50 122
1310 / 2888	SL		665	660	655	650	645	645	620	585
	2000	610	655	650	645	640	635	630	595	555
	4000	1219	645	640	635	630	620	605	565	525
	6000	1829	635	630	620	615	605	580	540	500
	8000	2438	620	615	605	600	590	550	505	465
	10000	3048	605	600	590	580	555	510	465	425
	12000	3658	590	580	570	560	520	480	435	395
	14000	4267	575	565	555	540	500	445	400	360
	16000	4877	560	550	520	470	405	360	320	280
	16400	4999	545	535	500	450	380	335	295	255
SL			690	685	680	675	670	665	645	605

Available tables:

Cruise Climb
Flaps UP

1310 kg
1280 kg
1200 kg
1100 kg

DA40 NG

240

Climb performance



■ Climb Performance

■ Tables show rate of climb

- Conversion to gradient:

$$Gradient [\%] = \frac{ROC [fpm]}{TAS [KTAS]} \cdot 0.98$$

- with TAS: „still air gradient“ !

- For operation without wheel fairings a climb rate decreased by 40 ft/min must be expected

DA40 NG

241

Time, Fuel, Dist to climb



Time, Fuel and Distance to Climb										
Flaps: UP					Power: 92% or max. 2100 RPM					
V _Y : 88 KIAS										
Weight [kg] / [lb]	Press. Alt. [ft]	Press. Alt. [m]	OAT [°C]	OAT [°F]	TAS [kt]	RoC [ft/min]	RoC [m/s]	Time [min]	Fuel [US gal]	Distance [NM]
1310 / 2888	SL		15	59	87	650	3.3	0	0.0	0
	2000	600	11	52	88	645	3.3	3	0.4	5
	4000	1219	7	45	90	645	3.3	6	0.9	9
	6000	1829	3	38	91	640	3.2	9	1.3	14
	8000	2438	-1	30	92	630	3.2	13	1.8	19
	10000	3048	-5	23	94	625	3.2	16	2.2	25
	12000	3658	-9	16	95	620	3.2	19	2.7	31
	14000	4267	-13	9	97	615	3.1	23	3.1	37
	16000	4877	-17	2	98	605	3.1	26	3.7	43
	SL		15	59	87	675	3.4	0	0.0	0

Available tables:

Flaps UP

1310 kg
1280 kg
1200 kg
1100 kg

DA40 NG

242

Time, Fuel, Dist to climb

- Time, Fuel, Distance to Climb:
 - Based on ISA, zero wind
 - Add 5% to time and fuel,
10% to distance
for each 10° increase of OAT
 - Method of calculation:
 - 1. get values for airfield altitude
 - 2. get values for for cruise altitude
 - Result: subtract 1 from 2

DA40 NG

243

True Air Speed

Cruise Performance															
Press. Alt. [ft] / [m]	Outside Air Temperature - [°C]														
	ISA-10			ISA			ISA+10			ISA+20			ISA+30		
	Pwr [%]	FF [US gal/h]	TAS [kt]	Pwr [%]	FF [US gal/h]	TAS [kt]	Pwr [%]	FF [US gal/h]	TAS [kt]	Pwr [%]	FF [US gal/h]	TAS [kt]	Pwr [%]	FF [US gal/h]	TAS [kt]
2000 610	92	8.3	134	92	8.3	136	92	8.3	137	92	8.3	138	92	8.3	140
	75	6.6	123	75	6.6	125	75	6.6	126	75	6.6	127	75	6.6	128
	60	5.1	112	60	5.1	113	60	5.1	114	60	5.1	115	60	5.1	116
	45	4.0	95	45	4.0	96	45	4.0	97	45	4.0	97	45	4.0	98
4000 1219	92	8.3	137	92	8.3	138	92	8.3	140	92	8.3	141	92	8.3	142
	75	6.6	126	75	6.6	127	75	6.6	128	75	6.6	129	75	6.6	131
	60	5.1	113	60	5.1	114	60	5.1	116	60	5.1	117	60	5.1	118
	45	4.0	96	45	4.0	97	45	4.0	98	45	4.0	98	45	4.0	99
6000 1829	92	8.3	139	92	8.3	141	92	8.3	142	92	8.3	144	89	8.0	143
	75	6.6	128	75	6.6	129	75	6.6	130	75	6.6	132	75	6.6	133
	60	5.1	115	60	5.1	116	60	5.1	117	60	5.1	118	60	5.1	119
	45	4.0	98	45	4.0	98	45	4.0	99	45	4.0	99	45	4.0	100
	92	8.3	142	92	8.3	143	92	8.3	145	92	8.3	146	89	8.0	146

Only one table for
1310kg

↓
Up to
16000 ft

DA40 NG

244

Landing distance

Landing Distance - Flaps LDG - 1310 kg / 2888 lb								
Weight:		1310 kg / 2888 lb			Flaps:		LDG	
V _{REF} :		77 KIAS			Power:		IDLE	
Runway: dry, paved, level								
Press. Alt. [ft] / [m]	Distance [m]	Outside Air Temperature - [°C] / [°F]						ISA
		0 / 32	10 / 50	20 / 68	30 / 86	40 / 104	50 / 122	
SL	Ground Roll	305	315	325	335	355	375	318
	15 m / 50 ft	620	650	670	680	720	760	650
1000 305	Ground Roll	315	325	335	350	370	395	327
	15 m / 50 ft	640	660	680	700	740	790	660
2000 610	Ground Roll	325	335	350	370	390	415	336
	15 m / 50 ft	650	670	690	730	770	810	670
3000 914	Ground Roll	335	350	365	385	410	435	345
	15 m / 50 ft	670	690	710	750	800	840	681
4000	Ground Roll	350	360	380	405	430	455	356

Available tables:

Flaps LDG

1310 kg
1280 kg
1200 kg
1100 kg

Flaps T/O or UP

1310 kg
1280 kg
1200 kg
1100 kg

↓
Up to
10000 ft

Headwind:

LDG distance minus 10% for each 20 kt

Tailwind:

LDG distance plus 10% for each 3 kt

Downhill slope:

Ground roll plus 10% for each 1% of slope

DA40 NG

245

LDG on wet or Grass RWY

	LDG roll	
Paved RWY WET	+ 15%	
Length of grass		Wet grass or soft ground
- 5 cm	+ 30%	additional + 15%
> 5 cm	min + 45%	

DA40 NG

246

Go around

Go-Around Climb Performance											
Flaps: LDG			Power: MAX								
V _{REF} :			77 KIAS at 1280 kg (2822 lb) and 1310 kg (2888 lb) 76 KIAS at 1200 kg (2645 lb) 72 KIAS at 1100 kg (2425 lb)								
Weight [kg] / [lb]	Press. Alt. [ft]	Press. Alt. [m]	Rate of Climb - [ft/min]								
			Outside Air Temperature - [°C] / [°F]								
			-20	-10	0	10	20	30	40	50	ISA
			-4	14	32	50	68	86	104	122	
1310 / 2888	SL		410	405	395	390	385	375	360	335	388
	2000	610	395	390	380	375	370	360	340	310	376
	4000	1219	380	375	365	360	350	340	315	285	364
	6000	1829	365	360	350	345	335	315	285		351
	8000	2438	350	345	335	320	310	280	250		336
	10000	3048	330	320	310	295	275	240			315
1280 / 2822	SL		425	415	410	400	395	385	370	345	400
	2000	610	410	400	395	385	380	370	350	320	387
	4000	1219	395	385	380	370	365	350	325	295	375
	6000	1829	380	370	360	355	345	325	295		361
	8000	2438	360	355	345	330	320	290	260		346

Available tables:

Flaps LDG

1310 kg
1280 kg
1200 kg
1100 kg

DA40 NG

247

Glide

Best gliding Flaps UP, prop windmilling	88 KIAS
1:9,7	1,59 NM / 1000 ft
Without wheel fairings:	
1:9,4	1,54 NM / 1000 ft

DA40 NG

248

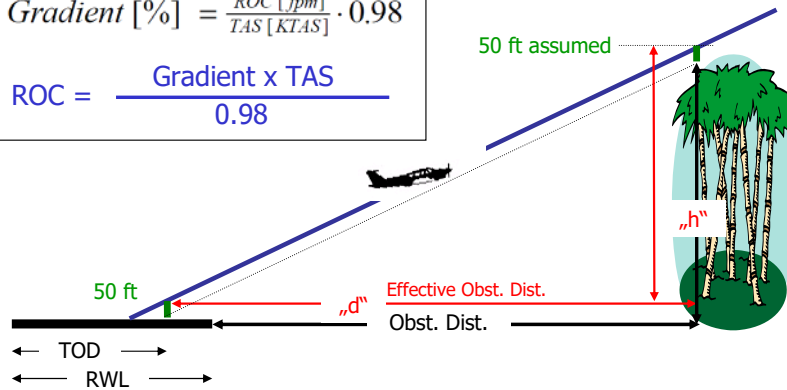
Obstacles

$$„d“ = (RWL + Obst. Dist.) - TOD$$

$$Gradient = („h“ / „d“) * 100$$

$$Gradient [\%] = \frac{ROC [fpm]}{TAS [KTAS]} \cdot 0.98$$

$$ROC = \frac{Gradient \times TAS}{0.98}$$



DA40 NG

249

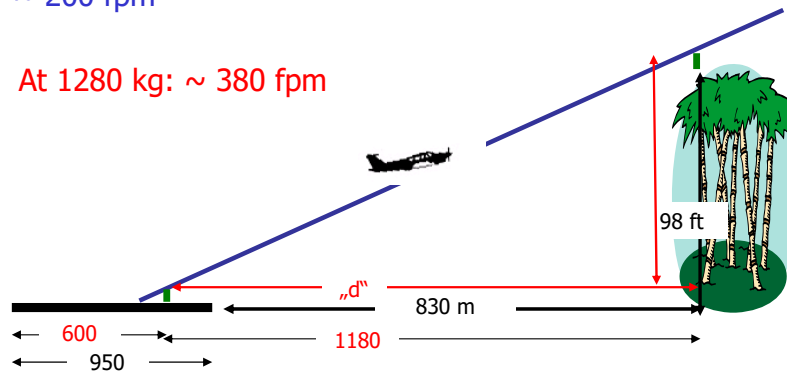
Vöslau RWY 31

$$„d“ = (950 + 830) - 600 = 1180$$

$$Gradient = (30 / 1180) * 100 = 2,54 \%$$

~ 200 fpm

At 1280 kg: ~ 380 fpm



DA40 NG

250



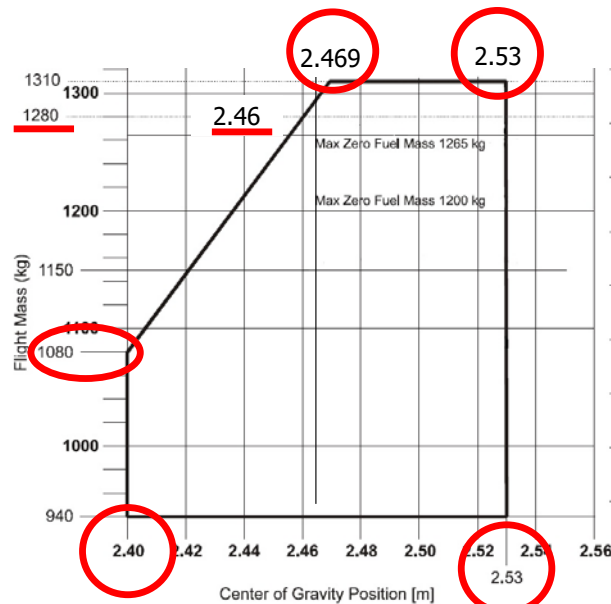
251

Empty Mass

- Empty Mass includes:
 - Equipment as per Equipment Inventory
 - Brake fluid
 - Coolant fluid
 - Gear oil
 - Engine oil
 - Unusable fuel (2 x 1,0 USG)

252

Center of Gravity Envelope



DA40 NG

253

© Diamond Flight Training

Compiled by Peter Schmidleitner

Moment Arms

Item		Lever Arm (m)
Front seats		2.30
Rear seats		3.25
Wing tanks		2.63
Standard baggage		3.65
Baggage tube		4.32
Short baggage extension		3.97
Extended baggage	FWD	3.89
	AFT	4.54

DA40 NG

254

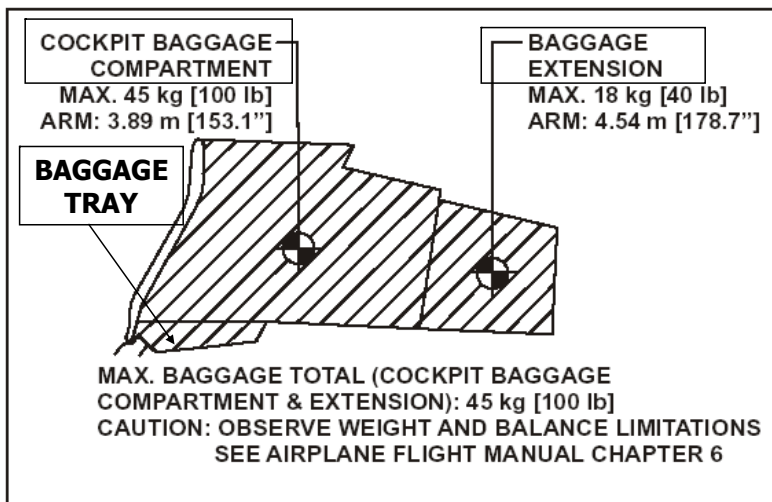
© Diamond Flight Training

Compiled by Peter Schmidleitner

Baggage Compartments



Option „Baggage Tray + Extended Compartment“



DA40 NG

255

DIAMOND FLIGHT TRAINING

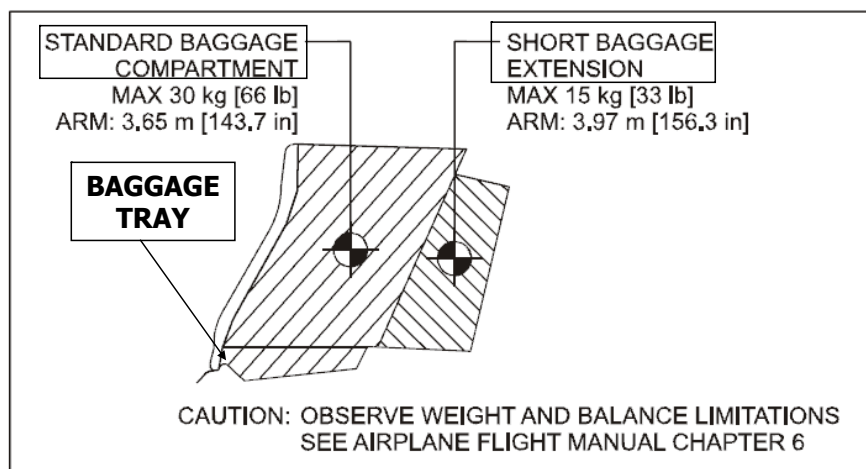
© Diamond Flight Training

Compiled by Peter Schmidleitner

Baggage Compartments



Option „Short Baggege Extension“



DA40 NG

256

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

M&B Calculation

	Lever arm	Mass (kg)	Moment (kgm)
Empty mass	2.42	900	2180.8
Front seats	$170 \times 2.30 = 391.0$		391.0
Rear seats	$80 \times 3.25 = 260.0$		260.0
Fwd baggage	3.89	10	38.9
Aft baggage	4.54		
Zero Fuel Mass		1160	2870.7
Fuel ~20 USG	2.63	60	157.8
Total TKOF Mass		1220	3028.5

DA40 NG

257

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

M&B Calculation

	Lever arm	Mass (kg)	Moment (kgm)
Empty mass	2.42	900	2180.8
Front seats	2.30	170	391.0
Rear seats	3.25	80	260.0
Fwd baggage	3.89	10	38.9
Aft baggage	4.54		
Zero Fuel Mass	2.47		$2870.7 : 1160 = 2.47$
Fuel	2.63	60	157.8
Total TKOF Mass	2.48		$3028.5 : 1220 = 2.48$

DA40 NG

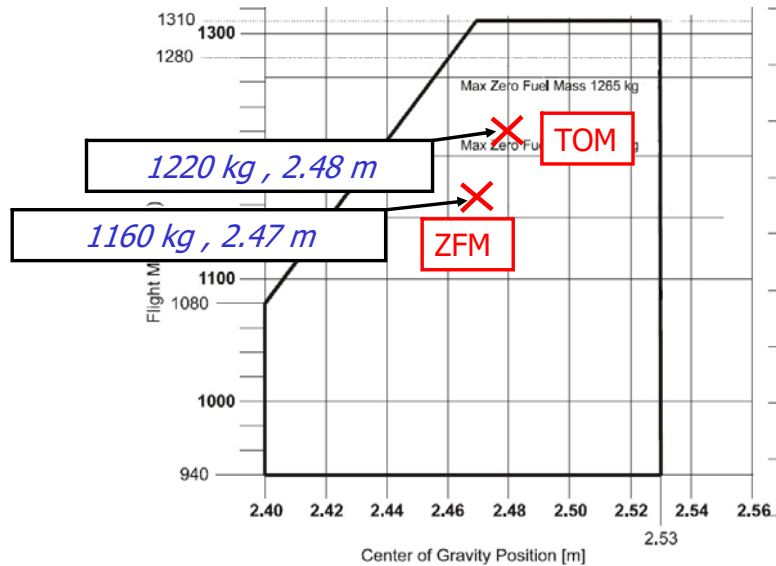
258

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Center of Gravity Envelope



DA40 NG

259

Intentionally blank

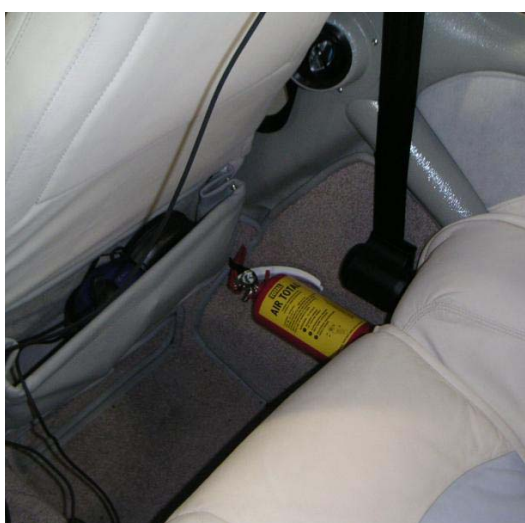
DA40 NG

Diamond DA40 NG



261

Emergency equipment



DA40 NG

262

Emergency Exit



DA40 NG

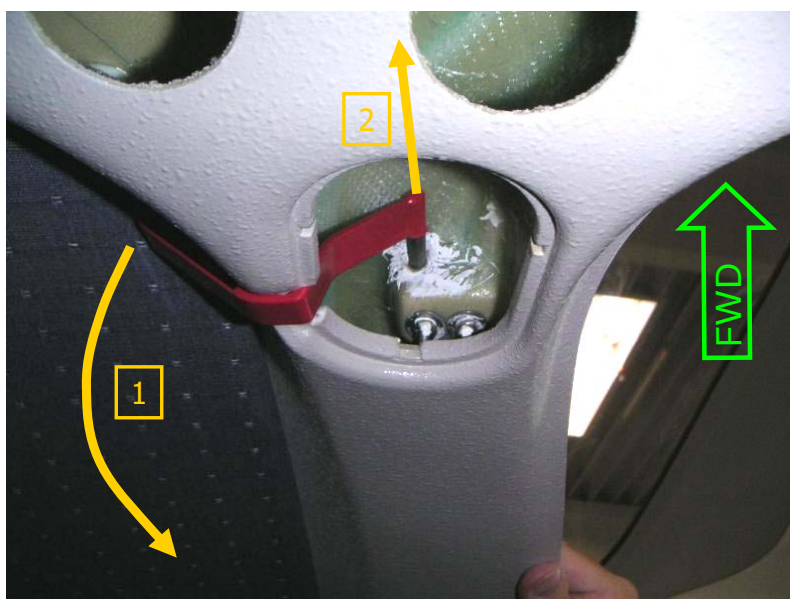
263

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Emergency Exit



DA40 NG

264

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Egress Hammer



DA40 NG

265

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Intentionally blank

DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG

Kinds of Operation Equipment List (KOEL)



267

Kinds of Operation Equipment List KOEL

	For daytime VFR flights	In addition for night VFR flights	In addition for IFR flights
Flight & navigation instruments	- Airspeed indicator - Altimeter - Magnetic compass 1 headset, used by pilot in command	- Vertical speed indicator (VSI) - Attitude gyro - Turn & bank indicator - Directional gyro - VHF radio (COM) - VOR receiver - Transponder (XPDR) - GPS receiver (part of G1000, if installed) - Second headset (if PM 1000 intercom is installed)	- Second airspeed indicator (on PFD and backup, if G1000 is installed) - Second altimeter - Second attitude gyro (on PFD and backup, if G1000 is installed) - Second VHF radio (COM) - VOR-LOC-GP receiver - Second GPS receiver (part of G1000, if installed)

DA40 NG

268

Kinds of Operation Equipment List KOEL



	For daytime VFR flights	In addition for night VFR flights	In addition for IFR flights
Engine instruments	- Fuel qty. - Oil press. - Oil temp. - Coolant temp. - Coolant level indicator - Gearbox temp. - Load - Prop. RPM - Fuel temp. left & right tank - Fuel flow - Fuel pressure warning - ECU A/B Caution	- Ammeter - Voltmeter	

DA40 NG

269

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Kinds of Operation Equipment List KOEL



	For daytime VFR flights	In addition for night VFR flights	In addition for IFR flights
Lighting		- Position lights - Strobe lights (anti collision lights) - Landing light - Instrument lighting - Flood light - Flashlight	
Other operational minimum equipment	- Stall warning system - Alternate means for fuel quantity indication (see Section 7.9) - Safety belts for each occupied seat - Airplane Flight Manual	- Pitot heating system - Alternate static valve	Emergency battery (for backup attitude gyro and flood light)

DA40 NG

270

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Kinds of Operation Equipment List KOEL



Additional minimum equipment
for the intended operation
may be required by
national operating rules
and also depends on the
route to be flown.

DA40 NG

271

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner



Intentionally blank

DA40 NG

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG



273

Scheduled maintenance

- Every
 - 100 hours
 - 200 hours
 - 1000 hours
- Annually

274

Unscheduled maintenance

- Hard landings
- Propeller strike
- Engine fire
- Lightning strike
- Occurrence of other malfunctions and damage

DA40 NG

275

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Refuelling



DA40 NG

276

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

De-icing

- Approved de-icing fluids:
 - Kilfrost TKS 80
 - Aeroshell Compound 07
 - Any AL-5 (DTD 406B)
- Procedure:
 - Remove snow with brush
 - Spray de-icing fluid
 - Wipe dry

DA40 NG

277

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Parking

- Short term parking:
 - Parking brake ON, Flaps UP
- Extended or unattended parking, unpredictable wind:
 - Mooring or placing in hangar
- Outside parking at OAT below -38° C:
 - Water / Coolant mixture ratio 40% to 60%
- If battery heating system installed:
 - Recommended to be used if OAT below 0° C

DA40 NG

278

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Control surfaces gust lock



DA40 NG

279

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Mooring



DA40 NG

280

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Mooring



DA40 NG

281

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Tow bar



DA40 NG

282

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Diamond DA40 NG



283

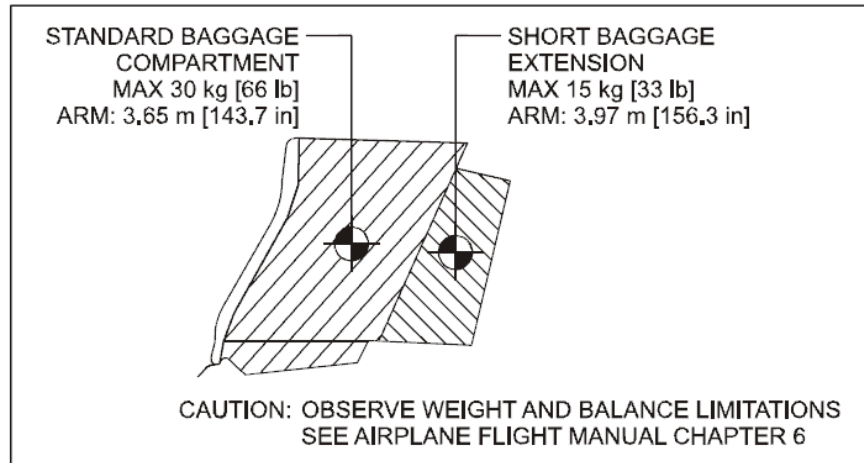
RACC:

Recirculating Air – Cabin Cooling

284

Installation

Behind the baggage compartment,
therefore only „Short“ baggage compartment



DA40 NG

285

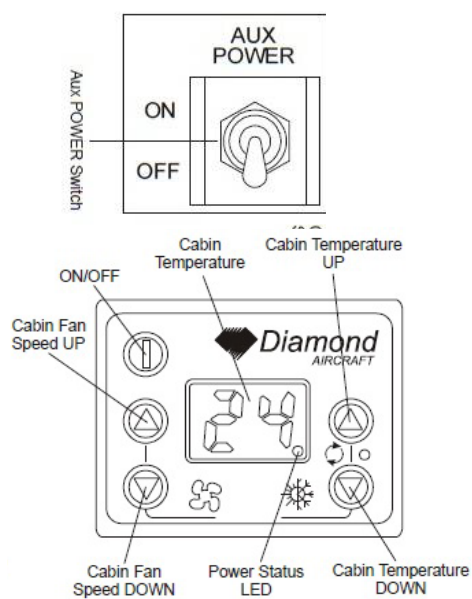
RACC Elements

- AUX Power switch + Control panel
 - on center console
- Central unit
 - aft of baggage compartment
- Additional alternator
 - engine compartment, belt driven from propeller shaft
 - Not connected to the electrical system!

DA40 NG

286

RACC Controls



DA40 NG

287

© Diamond Flight Training

Compiled by Peter Schmidleitner

Additional Alternator



DA40 NG

288

© Diamond Flight Training

Compiled by Peter Schmidleitner

Additional Alternator



DA40 NG

289

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

RACC Concept

- Cabin air is circulated through the RACC and thereby cooled
- Cabin air and outside air is also needed for the operation of the RACC

DA40 NG

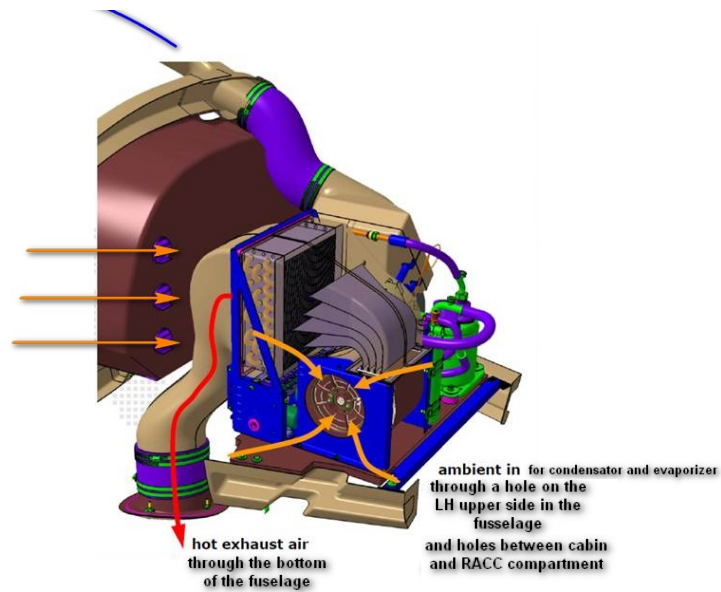
290

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

RACC Central Unit



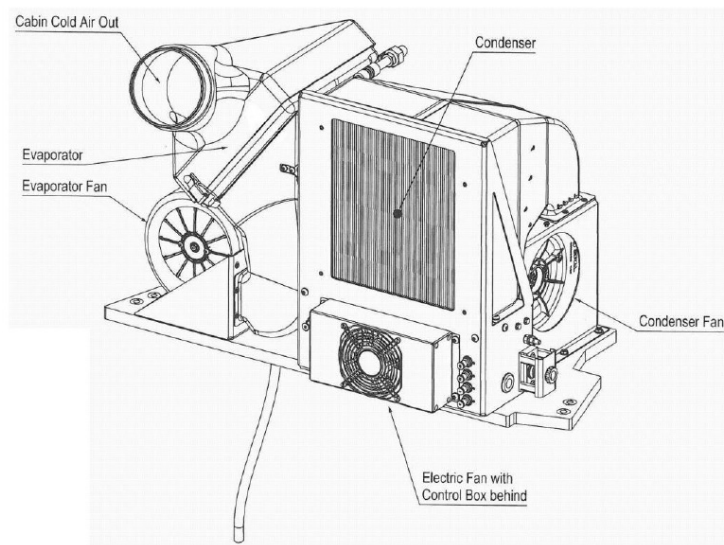
DA40 NG

291

© Diamond Flight Training

Compiled by Peter Schmidleitner

RACC Central Unit



Front and left side

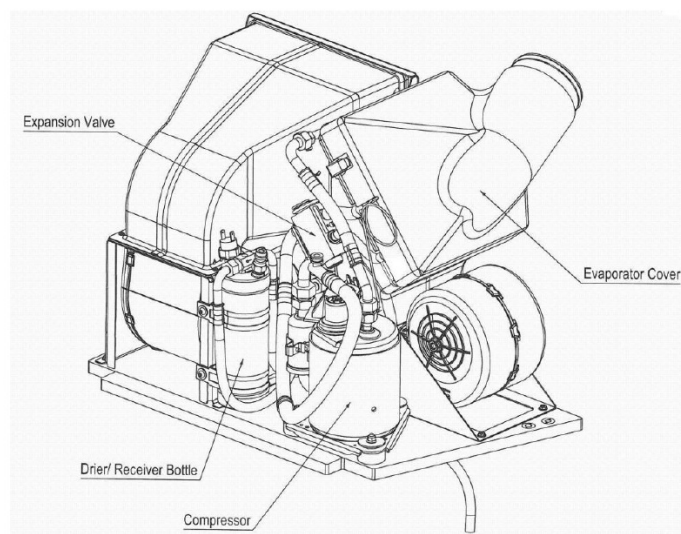
DA40 NG

292

© Diamond Flight Training

Compiled by Peter Schmidleitner

RACC Central Unit

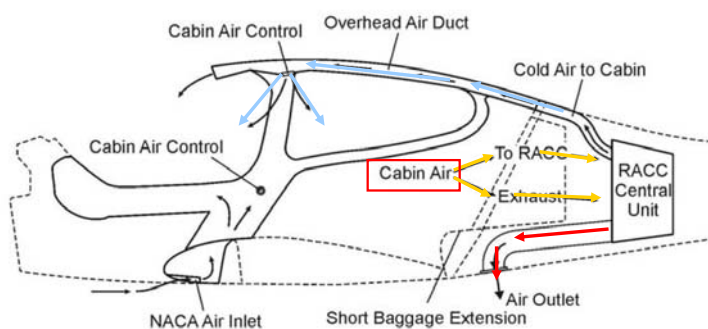


DA40 NG

Back and right side

293

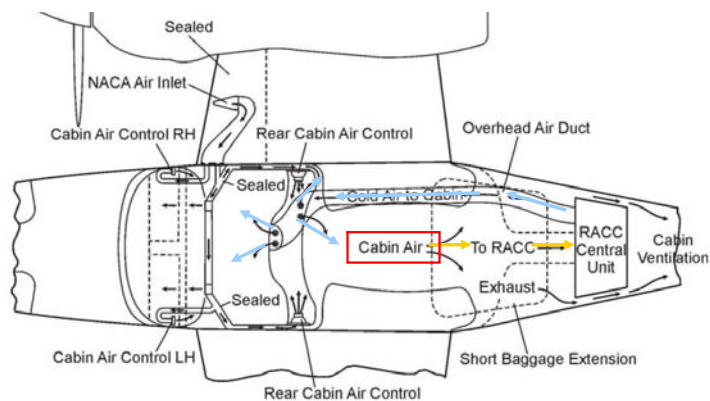
RACC Schematic



DA40 NG

294

RACC Schematic



DA40 NG

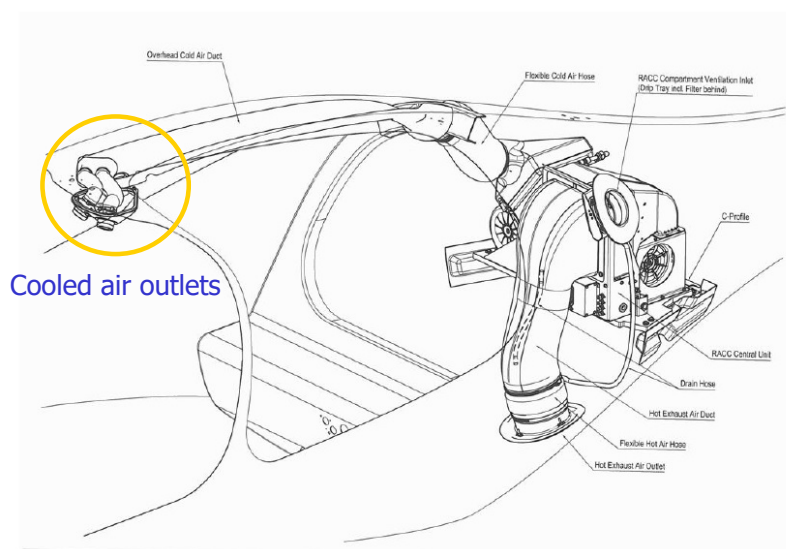
295

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

RACC Installation



DA40 NG

296

© Diamond Flight Training

DIAMOND FLIGHT TRAINING

Compiled by Peter Schmidleitner

Cooled air outlets



DA40 NG

297

RACC Limitations

- AUX POWER switch and RACC must be OFF
 - in all Emergencies
- during Abnormal Operating Procedures
- at OAT below 10° C (50° F)
- above 10.000 ft MSL
- when ADF system is used for navigation
 - RACC effects accuracy of the ADF system and the WX 500 Stormscope
- Minimum flight mass: 988 kg

DA40 NG

298

RACC Operation



- On ground with External Power:
 - AUX POWER: check OFF
 - ELECTRIC MASTER: check OFF
 - ENGINE MASTER: check OFF
 - Connect external power
 - RACC: ON
- On ground with engine running:
 - Keep RPM above 800
 - AUX POWER: ON
 - RACC: ON
- Power Off:
 - RACC: OFF (press button for 1 sec)
 - AUX POWER: OFF

DA40 NG

299

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

RACC Operation



In flight:

- switching RACC System ON:
 - AUX POWER: ON
 - RACC: ON
- switching RACC System OFF:
 - RACC: OFF
 - AUX POWER: OFF

DA40 NG

300

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Take-off

Take-Off Distance - Normal Procedure - 1310 kg / 2888 lb							
Weight: 1310 kg / 2888 lb				Flaps: T/O			
V _R : 67 KIAS				Power: MAX			
V ₅₀ : 72 KIAS				Runway: dry, paved, level			
Press. Alt. [ft] / [m]	Distance [m]	Outside Air Temperature - [°C] / [°F]					
		0 / 32	10 / 50	20 / 68	30 / 86	40 / 104	50 / 122

RACC ON:
ground roll increased by 30 m
TKOF distance increased by 40 m

DA40 NG

301

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Climb and Cruise

Cruise Climb - Flaps UP										
Flaps: UP						Power: 92% or max. 2100 RPM				
v _y : 88 KIAS										
b)			Rate of Climb - [ft/min]							
RACC ON: reduced by 25 ft/min										
Weight	[kg]	[lb]	-20	-10	0	10	20	30	40	50
			-4	14	32	50	68	86	104	122

RACC ON: reduced by 25 ft/min

Cruise Performance															
Outside Air Temperature - [°C]															
Press. Alt. [ft] / [m]	ISA-10			ISA			ISA+10			ISA+20			ISA+30		
	Pwr [%]	FF [US]	TAS [kt]	Pwr [%]	FF [US]	TAS [kt]	Pwr [%]	FF [US]	TAS [kt]	Pwr [%]	FF [US]	TAS [kt]	Pwr [%]	FF [US]	TAS [kt]
610	60	5.1	112	60	5.1	113	60	5.1	114	60	5.1	115	60	5.1	116
	45	4.0	95	45	4.0	96	45	4.0	97	45	4.0	97	45	4.0	98
	92	8.3	137	92	8.3	138	92	8.3	140	92	8.3	141	92	8.3	142

RACC ON: reduced by 4%

RACC ON: reduced by 4%

DA40 NG

302

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Moment Arms

Item	Lever Arm (m)
Front seats	2.30
Rear seats	3.25
Wing tanks	2.63
Standard baggage	3.65
Short baggage extension	3.97

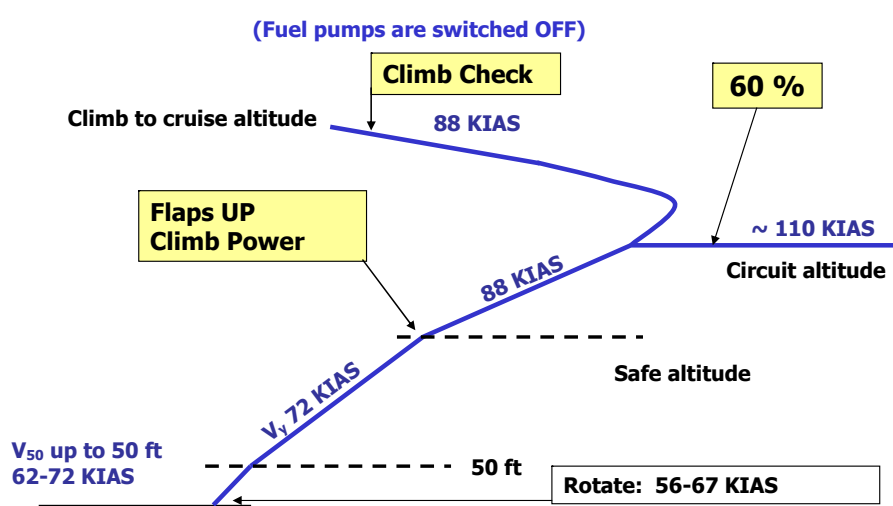
Intentionally blank

Diamond DA40 NG



305

TKOF Profile



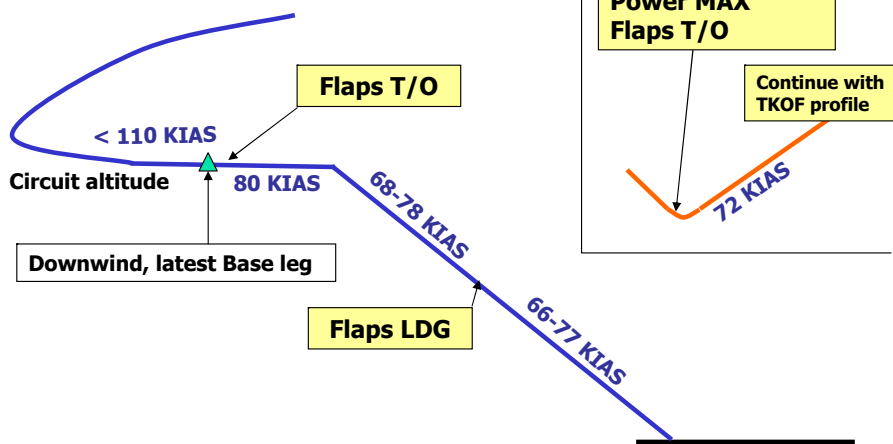
DA40 NG

306

LDG Profile

(Fuel pumps are switched ON)

Descent/Approach Check



DA40 NG

307

Checklists



The checklists are not part of the AFM, and they are not „officially Diamond factory-endorsed“.

They are published and used by „Diamond Flight Training“ and are also a recommendation for an „Operator Checklist“

DA40 NG

308